NETWORK NEUROSCIENCE

an open access journal

REVIEW

The neurobiological complexity of brain dynamics: The neurogliovascular unit and its relevance for psychiatry

Federico E. Turkheimer[1,2], Fran Hancock[1,2], Fernando E. Rosas[3,4,5,6], Anthony C. Vernon[7], Deepak P. Srivastava[7], Po-Wah So[2], and Andrea I. Luppi[8,9]

[1]The Institute for Human and Synthetic Minds, King's College London, London, UK.
[2]Department of Neuroimaging, Institute of Psychiatry, Psychology and Neuroscience, King's College London, London, UK
[3]Sussex Centre for Consciousness Science and Sussex AI, Department of Informatics, University of Sussex, Brighton, UK
[4]Centre for Complexity Science and Centre for Psychedelic Research, Department of Brain Science, Imperial College London
[5]Centre for Eudaimonia and Human Flourishing, University of Oxford
[6]Principles of Intelligent Behaviour in Biological and Social Systems (PIBBSS)
[7]Department of Basic & Clinical Neuroscience, Institute of Psychiatry, Psychology and Neuroscience, King's College London, London, UK
[8]Centre for Eudaimonia and Human Flourishing, Department of Psychiatry, University of Oxford, UK
[9]St John's College, University of Cambridge, Cambridge, UK

Keywords: Brain, Dynamics, Metastability, Multi-stability, Fractality, Development, Homeostasis, Excitation, Inhibition, Astrocyte, Microglia, Complexity, Mathematical modelling, Abductive inference

ABSTRACT

Modern neuroscience understands the brain as a complex system whose functional properties are determined by the myriad of interactions of its cellular components. In the difficult route to obtain better understanding of this organ, mathematical tools have become ever more effective, and we are now able to simulate the dynamics of brain signals from digital representations that have high fidelity to experimental data. However, these simulations are generated by inherently simple neuronal models that contain assumptions that, by necessity, can be at times far removed from the underlying neurobiology. Here, we develop an integrated view of brain dynamics that describes how the neurovascular units—made of neurons, capillaries, and glia—collectively determine the development of the functional properties of brain neuronal populations and maintain their homeostatic control. These interactions are instructed by genetic programs modulated by local environmental conditions. These complexities can be integrated into generative models and supply helpful analytics able to link the observable data to neurobiological entities if detailed brain mappings of molecular and cellular components are available. We further highlight how the development of these generative models could be relevant to the understanding of psychiatric conditions.

Citation: Turkheimer, F. E., Hancock, F., Rosas, F. E., Vernon, A. C., Srivastava, D. P., So, P.-W., & Luppi, A. I. (2026). The neurobiological complexity of brain dynamics: The neurogliovascular unit and its relevance for psychiatry. *Network Neuroscience*. Advance publication. https://doi.org/10.1162/NETN.a.588

DOI:
https://doi.org/10.1162/NETN.a.588

Received: 5 January 2026
Accepted: 17 April 2026

Competing Interests: The authors have declared that no competing interests exist.

Corresponding Author:
Federico E. Turkheimer
federico.turkheimer@kcl.ac.uk

Handling Editor:
Gustavo Deco

The MIT Press

PREMISE

Modern neuroscience understands the brain as a complex organ—an ensemble of billions of units that receive sensory signals from the environment and produce adaptive behavioral outputs as the result of a myriad of cellular computations taking place at various spatial and temporal scales (Barabási et al., 2023; Brynildsen et al., 2023; Turkheimer et al., 2022). However, scientific narratives that embrace this inherent complexity face a peculiar challenge, as efforts to build up from a comprehensive understanding of the biology and physiology of its cellular components often happen to need additional elements (e.g., computational models of particular subsystems) to properly account for higher levels of the organ, as scaling up from cells to populations and to systems is highly nontrivial (Bassett et al., 2018; Fornito et al., 2015; Pulvermuller et al., 2021). Fortunately, important insights about how the brain operates at mesoscopic and macroscopic levels have been recently gained by analyzing its functioning as a dynamic system (Brembs, 2021). In this framework, the four concepts of connectivity, computation, criticality, and coherence—the 4Cs—have been widely used to characterize brain's functionality in terms of its workings as a network, its capability to process information, its ability to switch rapidly across multiple states, and its capacity of transferring signals through dynamic modes of synchrony (Hancock et al., 2022; Luppi, Rosas, Mediano, et al., 2024; Sporns, 2022). These research efforts have led to the hypothesis that healthy brain function depends on the balancing of two tendencies: stable integration between brain areas for effective coordinated functioning and segregated processing within these same neuronal subsystems to utilize their functional specialization (Lord et al., 2017; Tononi et al., 1998). This unique signature has often been described as "metastability."

Mesoscopic:
It refers to a scale of measurement or analysis that sits intermediate between the microscopic (atomic/molecular) and macroscopic (bulk/visible) worlds.

Criticality:
In complexity science, it describes the state of a complex system poised at a phase transition boundary between order and disorder. At this "edge of chaos," systems exhibit optimal information processing, high adaptability, and scale invariance (power laws), meaning small triggers can cause massive, unpredictable ripple effects.

Metastability:
Refers to a transient state where a system remains trapped in a nonequilibrium configuration, mimicking stability before eventually transitioning to another state. It represents a delicate balance between order and chaos, allowing systems to be robust yet flexible, often acting as a "saddle-like" point that slows down change rather than trapping it permanently.

Small-worldness:
Refers to a network structure where most nodes are not neighbors but can be reached from any other node through a small number of steps, combined with high local clustering.

The neurogliovascular:
unit is a functional, multicellular complex in the brain composed of neurons, glial cells (astrocytes, microglia), and vascular cells (endothelial cells, pericytes). Together, they regulate cerebral blood flow, maintain the blood–brain barrier, and support brain homeostasis.

In previous work, we have reviewed the theoretical foundations and the empirical observations that have brought metastability to the fore of the neurosciences (Hancock et al., 2025). Complementing this work, here, we investigate the biological underpinnings of such dynamic signature, which, so far, have been lacking a consistent, unified account. Indeed, not only metastability may be achieved via different mechanisms of varying levels of abstraction but, as we will see, even computational models with biological plausibility can be heavily dependent on parameter values and initial conditions to achieve a dynamical behavior that successfully matches the observed brain dynamics (Castaldo et al., 2023; Ramezanian-Panahi et al., 2022; Touboul et al., 2011; van Vreeswijk & Sompolinsky, 1996). These facts lead to a natural question: How does the brain attain and subsequently stay into metastable regimes?

Here, we propose that metastability is supported by two key properties, the small-worldness of structural and functional elements and the excitation-inhibition balance. Crucially, we argue that these properties can only be properly understood when put together within the neurogliovascular system. We discuss the limited extent to which features of the neurogliovascular system are accounted for by current computational models, and avenues to improve this in the future. We also explore the potential role of neurogliovascular features on psychiatric conditions, and their potential to improve diagnosis and treatment.

PRELIMINARIES

When investigating the origins of metastability, a natural approach is to study computational models of brain activity, starting with basic simulations of individual neurons and then moving up-scale to neuronal populations and networks. In this process, because of the limited number of parameters one can include into a computational model before it becomes intractable, details are progressively lost and new macroscopic features may be introduced (Ramezanian-Panahi et al., 2022).

As between-neuron communication seems to be the principal determinant of brain dynamics, a good starting point for a computational model of the brain may be the generation of action potentials. The mechanism of this flow of ions was first explained by the influential Hodgkin-Huxley equations where the electrical current of the equivalent circuit is described by four differential equations that incorporate membrane capacity and the gating variables of the channels (Hodgkin & Huxley, 1990). However, when the same model has to be scaled up to population level, for realism, the multiple neuronal types present in the brain may need to be considered, which then requires some form of simplification for each neuron dynamic to avoid overparameterization. For example, Izhikevic and Edelman's (Izhikevich & Edelman, 2008) whole cortex model used up to 22 different pyramidal and inhibitory interneurons, but adopted a simplified mono-equation dynamic that only modeled firing rates (as opposed to ion currents). This model can demonstrate several emergent phenomena such as self-sustained spontaneous activity, chaotic dynamics, and avalanches, alongside oscillatory activities like those in the human brain. The model is, however, very sensitive to the firing-rate parameter settings and to the structural distribution of neuronal populations that, in this realization, was obtained from diffusion tensor images of adult subjects (please see next section) (Ramezanian-Panahi et al., 2022). Hence this modelling approach is faced with two fundamental questions: how does the brain get into its specific morphology and how does it maintain its homeostatic conditions.

Emergent phenomena:
Complex, large-scale patterns or behaviors arising from the simple, collective interactions of many smaller parts in a system. These properties are novel, often unexpected, and cannot be predicted merely by analyzing the individual components, illustrating that the whole is greater than the sum of its parts.

Populations and Networks

Contemporary neuroscience understands brain function in health and development, aging and disease, as emerging from widely distributed interactions of its units within and between spatial scales (cells, populations, anatomical regions, etc.) that may be conceptualized as networks (Papo & Buldú, 2024; Rubinov, 2024; Turkheimer et al., 2022). Following this perspective, a natural question is to inquire about how these networks give rise to metastability, and which are their key properties in allowing this.

Simulation studies have convincingly demonstrated that the emergence of metastable dynamics is contingent on the pattern of interactions between the modules of a system (Cabral et al., 2011; Friston, 1997; Pope et al., 2021; Shanahan, 2010; Strogatz, 2001). In particular, dynamic patterns of functional connectivity consistent with metastable dynamics have been observed to emerge when coupling (structural and functional) has "small-world" properties. Small-world networks (Watts & Strogatz, 1998) are sets of nodes, or vertices, that are well connected such that neighboring elements form highly connected clusters. These clusters are linked to the other distant ones by a smaller number of longer links so that the number of jumps between any two of the vertices in the network is small (e.g., they have short-path lengths). Note that small-world networks may emerge from underlying self-similar (e.g., fractal) morphology at finer scales (Bassett et al., 2006). A number of previous investigations have demonstrated self-similar properties of neuronal dendritic branches, which hold across a number of scales for each measurement (Caserta et al., 1995; Kim et al., 2011; Milosević & Ristanović, 2006; Morigiwa et al., 1989; Smith et al., 1996; Takeda et al., 1992; Wearne et al., 2005).

Fractals:
Complex, never-ending patterns that repeat themselves at different scales, known as self-similarity. These structures, where the part mirrors the whole, are formed by repeating processes in nature like growth, erosion, or branching.

Given the above, networks of anatomical connections derived from the human connectome data, that indeed possess small-world properties, have been incorporated within computational simulations of large-scale neural dynamics to demonstrate metastable properties of brain function (Cabral et al., 2011; Deco et al., 2009; Hellyer et al., 2015; Honey et al., 2009; Ofer et al., 2017; Páscoa dos Santos & Verschure, 2024). Empirical studies have then demonstrated that disruption to small-world brain networks is associated with neurodevelopmental and neurodegenerative disorders as well as brain injury (Hellyer et al., 2015; Liu et al., 2008; Pandit et al., 2013; Stam et al., 2007; Váša et al., 2015).

However, little is still known about the mechanisms driving such specific morphological architecture. Models put forward so far mirror the economy principle expressed by Ramón Cajal more than a century ago who proposed that neuronal morphology optimizes wiring (metabolic) costs (Ramón y Cajal, 1899); the underlying mechanism of network optimization has been characterized as evolutionary, hence stemming from genetic control either generalized across the brain or adapting to the local cellular composition (Akarca et al., 2021; Arnatkeviciute et al., 2021; Bassett & Bullmore, 2017; Bullmore & Sporns, 2012; Libby et al., 2017; Muldoon et al., 2016; Oldham et al., 2022; Vértes et al., 2012).

We propose here that the assumption of complete genetic control of the development of small-world neural networks is not very plausible (Bremermann, 1963; Duclos et al., 2019). Indeed, the number of genes in the genome correlates almost linearly with the number of cell types in the various organisms, but it is far too low to code for the regulation of the interactions across all their elements (Kauffman, 1991). Hence, a more likely assumption is that genetic programs may be activated or suppressed by local interactions, for example, cell-to-cell information exchanges via epigenetic, hormonal, metabolic, chemical, and electrical mechanisms that ultimately control cellular differentiation and biological function; this interconnected set of relation is generally referred to as the "physiome" and is described in detail in The Complexity of the Determinants of Brain Dynamics section (Bassingthwaighte, 1992; Duclos et al., 2019; Hunter & Borg, 2003; Lindenmayer, 1968). In other words, brain network structures emerge from the complex interactions between the genetic coding of each cell and the local functional interactions among cells; note that the term "interaction" is used here instead of the term "causation" as causation in complex systems very often works both ways and the elements are generally difficult to disentangle in their dynamic evolution. For more on this topic, please see Turkheimer et al. (2022).

Excitation-Inhibition Balance

A second key aspect of the emergence of brain dynamics is the combination of small-world networks with an appropriate balance between excitatory and inhibitory activities within the neuronal populations (Castaldo et al., 2023; Poil et al., 2012; van Vreeswijk & Sompolinsky, 1996), which we revise now.

Cellular units made by a pair of glutamatergic pyramidal (e.g., excitatory) and GABAergic (e.g., inhibitory) neurons constitute the canonical generators of brain signals (Turkheimer et al., 2015). The inhibitory action dampens spike-trains and morphs them into oscillations with frequency dependent on the feedback delay (Buzsáki & Wang, 2012) (Figure 1).

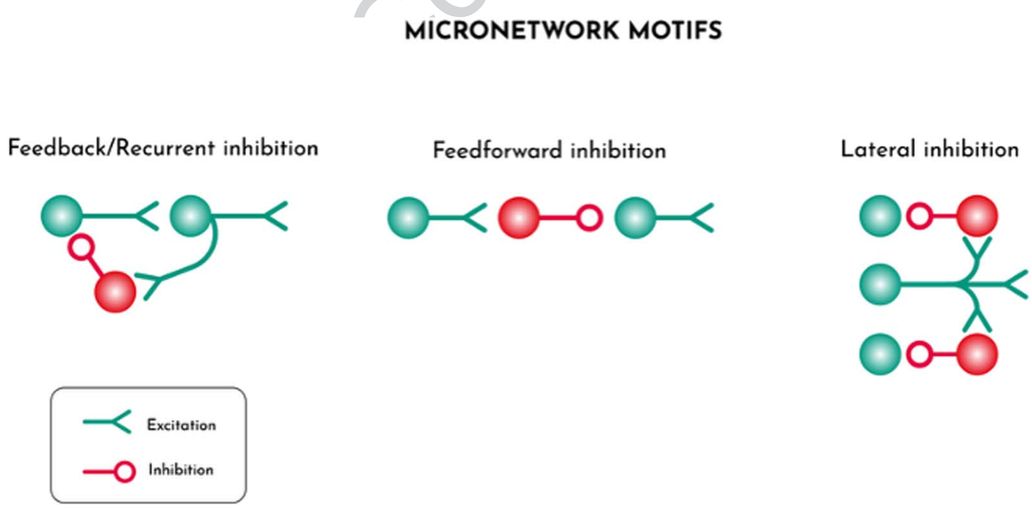

Figure 1. Alternative architectures of elementary feedback systems that enable the generation of fast frequency waves, the building blocks of brain functional connectivity. Excitatory units are constituted by pyramidal cells while inhibitory cells are GABA interneurons.

For example, parvalbumin basket cells are activated by the excitatory signal transmitted by the paired pyramidal neurons and feedback negatively on the nucleus of these units with a delay of ~20 ms, generating waves with a gamma power of 50 Hz (Börgers & Kopell, 2003; Wang & Buzsáki, 1996). Besides feeding back to the neurons they are coupled with, the same GABA interneurons also reach out to the inner part of the neuronal columns or project to the surrounding columns dampening the activities therein so that the traveling oscillations become three-dimensional wave fronts (Helmstaedter et al., 2009; Turkheimer et al., 2015). These oscillatory fronts are the binding mold that allows the synchronization of local neuronal ensembles but also, as they travel through distances in the brain, combine to generate slower waves that facilitate spatial synchronization across larger sections of the cortex (Turkheimer et al., 2015).

The basic cellular mechanisms described above support the use of mathematical oscillators as realistic models of brain functional units; these can be used at any scale, from the cellular to the regional to the population (Breakspear, 2017; Strogatz, 2001). Hence, one key aspect of the maintenance and propagation of signals in brain networks is the tight balance of excitation and inhibition within neuronal populations across scales. For example, in sensory cortical neurons, an increase in synaptic excitatory activity is followed closely by a concomitant increase in inhibitory conductance (Isaacson & Scanziani, 2011; Wu et al., 2011). A tight balance between synaptic excitation and inhibition is also observed during spontaneous network oscillations at rest (Isaacson & Scanziani, 2011; Okun & Lampl, 2008; Shu et al., 2003).

Excitation-inhibition (E/I) heterogeneity across the brain and its apportioning of neuronal excitatory and inhibitory populations can be included into generative models (Deco et al., 2021; Kong et al., 2021; Zhang et al., 2024). Homeostatic conditions across the whole synaptic functional repertoire can also be achieved by adding to the model a plasticity component, for example, the possibility of slight local variations in excitatory and inhibitory ratios (Agnes & Vogels, 2024; D'Amour & Froemke, 2015; Kirchner & Gjorgjieva, 2021; Litwin-Kumar & Doiron, 2014; Mapelli et al., 2016; Vogels et al., 2011). These models have been able to demonstrate self-tuning mechanisms that work on the system at rest but allow sufficient plasticity to remap sensory representations of new sets of stimuli (Poil et al., 2012; van Vreeswijk & Sompolinsky, 1996; Yang & La Camera, 2024).

However, these attempts to biological fidelity are achieved by rules that are dependent on initial conditions, such as an appropriate ratio of GABA versus glutamatergic neurons, and on the hand-tuning of model parameters; importantly, with a few exceptions (Agnes & Vogels, 2024; Kirchner & Gjorgjieva, 2021), they do not necessarily reflect cellular specific mechanisms.

From Small-World and E/I Balanced Networks to Metastability

The material above has reviewed two key aspects of the generative models used in the simulation of brain dynamics—that is, the structure of the networks (e.g., small worlds) and their homeostatic control (e.g., E/I balance). In this section, we demonstrate how these two properties are important determine the transient states of dynamic connectivity that we have previously defined as metastability.

In order to elicit metastability, let us first note that the network needs some form of modularity. To see this, let us consider a toy example with three nodes—denoted by A, B, C—generating oscillatory activity. Let's assume that B is the "hub" node, being connected to A and C, while A and C are not directly connected but, upon some general conditions, their connection to B allows an in-phase quick synchronization (Gollo & Breakspear, 2014; Sporns & Kötter, 2004). Under these conditions, the addition of any additional input into nodes A and C can easily "frustrate" the existing state of synchrony, so that the nodes A and C may then proceed to establish novel synchrony states using another hub node as pivot. The preferred state of B as a hub for A and C prevents immediate system-wide synchronization and allows the occasional return of A and C to the original functional state. This dynamic state of affair can be repeated across space and across scales wherever these motifs are implemented in a modular structure comprising a hub and two communities (Gollo & Breakspear, 2014; Sporns & Kötter, 2004). The resulting synchronized states equilibrate at frequencies that depend on the distances and the oscillatory states of the units of the motif (Gollo & Breakspear, 2014; Sporns & Kötter, 2004). This supports the idea that a small-world structure with distinctive neuronal hubs across scales may be suitable to induce metastable states to be formed across the swath of brain space.

At the same time, for the motifs to be reliably repeated across space and time, the neuronal masses should be able to maintain their oscillatory activity notwithstanding a wide envelope of potential synaptic inputs as well as plastic adaptations necessary to brain development and function. Theoretical work has determined that, to maintain such robustness, a stable excitation-inhibition ratio based on strong excitatory and inhibitory currents internal to the oscillatory unit and stronger than any external input may be required (Páscoa dos Santos & Verschure, 2024; van Vreeswijk & Sompolinsky, 1996, 1998).

THE COMPLEXITY OF THE DETERMINANTS OF BRAIN DYNAMICS

Building on the considerations outlined above, here, we posit that to build generative models that properly account for the brain's biology, we may need to consider the whole cellular repertoire of the tissue. This view poses substantial challenges, as each simulation is limited by the limited number of parameters that simulations can be consider at any one time. Accordingly, it is crucial to keep in mind that neural models bring with them a set of conditions related to non-neuronal determinants—which holds true even when neuronal only models are considered. In the sequel, we provide evidence that:

A. The development of small-world neural networks can be understood in the context of their interactions with the development of vascular networks.
B. Homeostasis, and specifically a tight E/I balance, relies heavily on the role of glia, and particularly, astrocytes.

To advance the argument, we first briefly review the main neurobiological phases and determinants of brain development and particularly of neural and vascular networks, their mutual connections, as well as their interactions with glia. After this is done, the next section highlights the potential relevance of these aspects to psychiatric disorders.

The Development of Neural Networks in the Human Brain

The formation of branched structures that ultimately shape neuronal networks relies on processes of arborization and polarization that take place during neurogenesis in the development of each neuron (Guidolin et al., 2019). As reviewed below, such processes rely on the concerted switching on and off of genetic programs based on other cellular interactions.

Brain development proceeds in overlapping phases: the making of the brain cells, cell migration, axons and dendrites growth, synaptic development and linkage with other neuronal

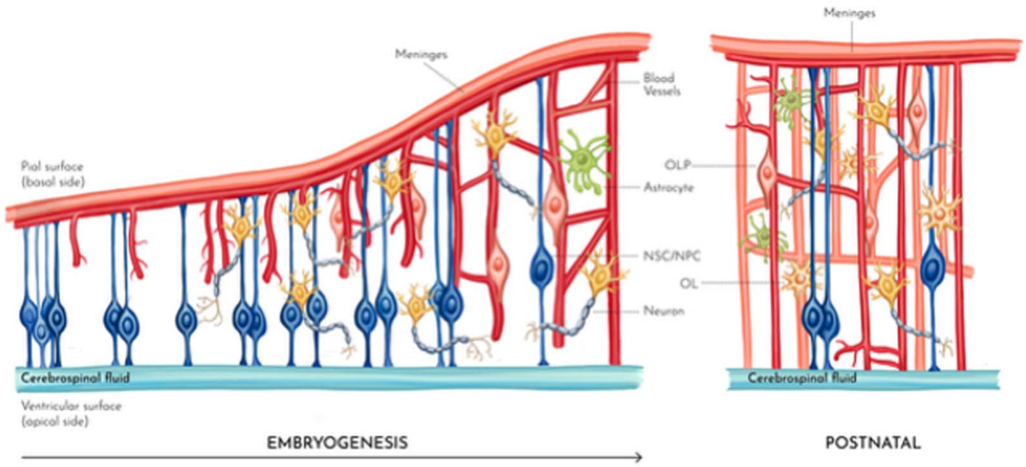

Figure 2. From left to right, the various stages of cortical formation during embryogenesis up to the postnatal brain. Neural stem/progenitor cells (NSCs/NPCs) start a first differentiation stage mostly converting into radial glial cells (RGCs) that make the scaffolding starting from the ventricular up to the pial surface while novel blood vessels grow inwards. Local metabolic, ionic, and molecular conditions then drive the generation and migration of glial and neuronal cells and their connections to the developing vascular networks. Finally, the same local conditions promote the differentiation and maturation of oligodendrocytes (OL) from their progenitor cells (OLP) and axonal myelination.

cells, synaptic maturation and pruning, glia-genesis, and myelination (Shonkoff et al., 2000) (Figure 2).

The brain and spinal cord arise from a set of cells on the dorsal part (e.g., neural plate) of the developing embryo. The dividing cells proliferate on each side of the plate along its length and their folding forms the neural tube. In the tube, three contained enlargements are formed from which the various parts of the brain are generated; the anterior portion gives rise to the cerebral hemispheres, the middle section to the midbrain, the posterior to the brain stem, the cerebellum, and the spinal cord and peripheral nerves. Under the control of regulatory genes, the NSCs/NPCs start a first differentiation stage mostly converting into RGCs; these cells are polarized and stretch radially from the early ventricles to the pia becoming the pillars of the expanding cortex. This is the scaffolding that guides the next phase when neurons travel to their appropriate layer in the neural columns (Shonkoff et al., 2000). This process stretches until the 15th gestational week after which astrocytes and oligodendrocytes commence their proliferation from RGCs and migration to support the developing neural columns. Note that, differently from the other glia, microglial cells appear very early in gestation (from Week 8) and then distribute across the developing cortex (Menassa & Gomez-Nicola, 2018).

Once the nerve cells are formed and finish migrating, they rapidly extend axons, often over relatively long distances, as well as dendrites and begin to form connections with other nerve cells; these connections are called synapses. This process continues well into the childhood years. There is evidence in many parts of the nervous system that the stability and strength of these synapses are largely determined by synaptic activity, and hence are dependent on environmental stimuli. As nerve cells form axons, accompanying oligodendrocytes wrap them with an insulating coating, myelin, that increases conduction velocity.

Axon guidance is a complex process whereas guidance receptors expressed at the tips of motile axons interact with guidance cues; these cues present with both positive (attraction) and negative (repulsion) feedback systems that are locally switched on and off in temporal sequelae along the path to avoid the formation of loops (Zang et al., 2021). Environmental clues also drive the arborization of axons, the process that extend multiple branches from the tip end to form terminal arbores (Kalil & Dent, 2014). The most important of these interactions are those with the vascular network.

The brain is one of the most energy consuming organs whose metabolism is mostly reliant on oxidative metabolism and holds minimal glycogen reserve; hence, it requires a constant supply of oxygen and glucose (Lord et al., 2013; Oz et al., 2007). The distribution of synapses, where most of the energy budget is spent with 50% used on postsynaptic glutamate receptors, must then spatially match the blood vessel area that supplies glucose and oxygen (Howarth et al., 2010, 2012).

The close interaction between neural and vascular networks in the brain is well documented and has even gained a specific label by being called the “neurovascular unit” (Eichmann & Thomas, 2013; Iadecola, 2017).

Hence, we propose that the small-world brain morphology of neural networks is based on the small-world morphology of the neurovascular unit. It is relevant to note that other organs that have very high metabolic demand, if not higher than the brain, such as the kidneys, heart, and liver, have all been shown to possess branching morphology and that this is not the case for other organs with lower metabolic needs (e.g., adipose tissue and skeletal muscle) (Captur et al., 2017; Flanagan et al., 2021; Smith & Trachtenberg, 2007; Wang et al., 2010). We develop this argument further in the following section.

The Development and Cross-Talk of Vascular Networks

Contrary to neural networks, the mechanistic bases of small-world development and morphology are well understood and modeled as fractal biology when it comes to the vascular system (Bassingthwaighte, 1992). The morphogenesis of vascular networks is based on the proliferative activity of equipotent tips that branch out to respond to metabolic demands, for example, hypoxia, and become proliferatively inactive when the latter are quenched and/or they enter an already oxygenated territory; while expanding, they also suppress the sprouting of neighboring tips (Hannezo et al., 2017). Hence, vascular branched structures develop as a self-organized process, rely on simple local rules, and do not require a deterministic sequence of genetically programmed events (Cardy & Täuber, 1996; Hannezo et al., 2017).

Specifically, the process of vascular branching is based on two elegant feedback systems. During development, tissue expansion causes hypoxic conditions that are sensed by the local astrocytic population that releases the vascular endothelial growth factor (VEGF). VEGF release stimulates the sprouting of endothelial cells that expand the vascular network and bring oxygenated blood until VEGF expression is switched off (Gerhardt et al., 2003). For such branching system to work, only a few endothelial cells need to be selected as tips while the others trail these and form tubes; this system is controlled via a form of lateral inhibition that blocks sprouting from adjacent cells via the Notch signaling pathway. (Roca & Adams, 2007).

Importantly for what is purported here, there is a close cross-talk between the vascular and neural networks that share signals through known axon and blood vessel guidance molecules: the Neuropilin receptors and their Semaphorin and VEGF ligands, and Netrins and Slits ligands and their receptors (Carmeliet & Tessier-Lavigne, 2005; Eichmann & Thomas, 2013; Iadecola, 2017; Jones & Li, 2007). A few models have been proposed to demonstrate the development of brain vasculature (Ii et al., 2020; Karch et al., 1999; Peyrounette et al., 2018; Shipley et al., 2020); Kumar and colleagues have proposed the first synthetic formulation of the interaction between growing vascular and neuronal networks (Kumar et al., 2022).

The Control of Excitation-Inhibition Ratio Balance

Approximately one in six neurons in the adult neocortex are inhibitory GABAergic interneurons, and this ratio is relatively stable across development but not across layers—with higher densities observed in Layers 2 and 5 (Beaulieu, 1993; Goulas et al., 2021; Meyer et al., 2011; Sahara et al., 2012). GABA interneurons are a highly heterogeneous group of neurons with diverse morphologies and biochemical and physiological properties (Lim et al., 2018). These features are acquired during development through the implementation of specific transcriptional programs that are intrinsically encoded and switched by local cues or driven by local interactions with the microenvironment (Lim et al., 2018). In particular, the laminar distribution, phenotype, connectivity, and even survival of interneurons are determined by signals provided by pyramidal cells (Hevner et al., 2004; Lodato et al., 2011; Miyoshi et al., 2010; Pla et al., 2018; Southwell et al., 2012; Ye et al., 2015). The maturation of interneurons and of their connections to pyramidal neurons drives the early appearance of local networks and gamma oscillations (Gestation Week 24) until full maturation of brain dynamics up to 2 years from birth (Wu et al., 2024).

Interneurons have well established role in inducing long-term plasticity at excitatory synapses (Wigström & Gustafsson, 1985); at the same time, as the result of interaction between excitatory and inhibitory synapses, excitatory transmission modulates inhibitory synaptic plasticity (Belan & Kostyuk, 2002). These cross-modulatory effects are well captured by a few of the mathematical models proposed (Pulvermuller et al., 2021).

However, current mathematical models do not account for glial cells (Fields et al., 2014; Ramezanian-Panahi et al., 2022). Glial cells are the most abundant (90%) cells in the human brain and are tightly integrated into neural networks. As already reviewed in previous sections, they have a key role in the development of vascular and neural networks and, in the mature brain, they are responsible for homeostatic control. They sense and shuffle major ions, remove and catabolize neurotransmitters, and supply neurons with neurotransmitter precursors; crucially they maintain metabolic homeostasis by metabolizing glucose and supplying neurons with energy substrates, mostly lactate, and by synthesizing glycogen (Verkhratsky & Nedergaard, 2018). With their perisynaptic sheaths, astrocytes enwrap the whole synapses and regulate synaptic maturation, maintenance, and extinction (Verkhratsky & Nedergaard, 2014).

Moreover, astrocytes use their large array of transporters to control the dynamics of neurotransmitter concentration in the synaptic cleft and modulate a wide array of synaptic responses (Marcaggi & Attwell, 2004). For example, astroglial release of glutamate may results in both inhibition or potentiation of evoked and spontaneous excitatory or inhibitory postsynaptic currents as well as modulation of long-term potentiation or long-term depression (Araque et al., 1998; Han et al., 2012; Jourdain et al., 2007; Kang et al., 1998; Liu et al., 2004; Min & Nevian, 2012; Navarrete et al., 2012; Santello et al., 2011; Santello et al., 2012). Conversely, astroglial GABA transporters have a key role in controlling concentration of GABA in the synaptic cleft and hence control GABA tone (Kersanté et al., 2013).

THE RELEVANCE OF THE NEUROGLIOVASCULAR UNIT TO PSYCHIATRIC DISORDERS

Unsurprisingly, abnormalities in these vascular and glial compartments have been involved in the pathophysiology of psychiatric disorders (Clark-Raymond & Halaris, 2013; Collignon et al., 2024; Lee & Kim, 2012; Palmer et al., 2000). VEGF plays a role not only in cerebrovascular development but also in neurogenesis in the adult brain and its dysregulation has been consistently associated with major depressive disorder (Clark-Raymond & Halaris, 2013; Collignon et al., 2024; Lee & Kim, 2012; Nunes et al., 2022; Palmer et al., 2000; Xie et al., 2017). Signaling pathways that we have reviewed above as key to microvascular development have also been linked to developmental disorders, schizophrenia, bipolar disorder, and autism (Hoseth et al., 2018; Wang et al., 2023; Yang et al., 2024). Microvascular abnormalities measured by retinal imaging in fact are a consistent finding in patients with schizophrenia (Katsel et al., 2017; Meier et al., 2013; Wang et al., 2023). A study of human induced pluripotent stem cells-derived neural stem cells of patients with schizophrenia has demonstrated impaired angiogenesis (i.e., the process of development of new blood vessels) due to defective cross-talk between neural stem cells and endothelial cells during the formation of the neurovascular niche (Casas et al., 2018).

Angiogenesis:
The formation of new blood vessels from preexisting vasculature, crucial for developing the brain's vascular network.

However, morphological information about abnormalities of the brain microvasculature in psychiatric cohorts is scant. In particular, total cerebral blood volume has been reported as below normal levels in patients with schizophrenia (Brambilla et al., 2007) and this seems to be the result also of underdeveloped small arterial (pial) and arteriolar vessels (Hua et al., 2017). It is worth noting that MRI methods to quantify the density of both arterial and venous micro-vessels are available (Balchandani & Naidich, 2015; Hua et al., 2011; Sarabi et al., 2024).

The relevance of excitation-inhibition imbalance in psychiatric disorders has been widely documented with findings in clinical cohorts with schizophrenia, autism spectrum disorder, as well as mood disorders (see Turkheimer et al., 2015, for a review) and the estimation of local excitation-inhibition balance is now an important effort in the development of analytics for brain functional imaging data (Colombo et al., 2019; Deco et al., 2014; Gao et al., 2017; Páscoa dos Santos & Verschure, 2024; Podvalny et al., 2015; Waschke et al., 2021; Zhang et al., 2024). Direct in vivo quantification of glutamate (Glu), and γ-aminobutyric acid (GABA) steady-state concentrations can be achieved using magnetic resonance spectroscopy (MRS) (Rae, 2014). However, it is still challenging to apportion the concentrations of these neurotransmitters to the underlying cellular mechanisms, for example, neuronal or glial (Lea-Carnall et al., 2023). Initial approaches included the use of metabolic activity in astrocytes as a proxy of cycling Glu given its high metabolic costs (Rothman et al., 2003); only very recently, MRS has been used to quantify time-resolved metabolic responses to stimuli on much shorter timescales of seconds to minutes, so-called functional MRS (Jelen et al., 2018), and these signals have allowed the attribution of Glu and GABA concentrations to cellular compartments (Lea-Carnall et al., 2023). Moreover, information on neuro-glial network microstructures can potentially be obtained from diffusion-weighted (DW) MRS (Palombo et al., 2018). Conventional diffusion MRI utilizes water molecules to probe and quantify microstructure whereas DW-MRS provides compartment specific information using glia compartment-specific metabolites as probes, for example, myoinositol for the astrocytic compartment.

THE NEUROGLIOVASCULAR UNIT AND GENERATIVE MODELS OF BRAIN NETWORKS

The previous sections have summarized how proper cerebrovascular development and neurogliovascular unit assembly are essential for brain growth and function, during development and throughout life, ensuring the continuous supply of nutrients and oxygen. According to the recently proposed model of supply-limited (as opposed to demand-based) metabolism, brain development, homeostasis, as well as brain evolution can be best understood as determined by the deliverable mass of metabolic substrates, oxygen, and glucose (Herculano-Houzel & Rothman, 2022; Luppi, Rosas, Noonan, et al., 2024); hence, angiogenesis plays a key role brain in modern theories of developmental disorders (Carrier et al., 2020). This supports the view that the inclusion into computational models of components of the neurogliovascular unit is relevant to contemporary neuropsychiatry; these models could then be tested against newly generated data made available by novel imaging technologies as reviewed above (Figure 3).

However, the relevant question that remains is how the complexity of neurovascular energetics can be captured by stylized models of brain networks development that may be computable and yet realistic. For this purpose, we propose an example using a specific class of computational models, the so-called generative models of brain network development as reviewed in Vértes (2023). Note that this selection is not prescriptive but is preferred here given the focus on vascularization and energetics. Further to generative modeling, there is a large body of work on computational modeling of brain development that explicitly considers biophysical properties and processes such as neurogenesis (Beul et al., 2018), neurite growth (Van Ooyen et al., 1995), and cell division (Kerstjens et al., 2022). Equally, computational models have been used to model GABA/glutamate homeostasis (Naskar et al., 2021) and glial/neuronal interactions (Nadkarni & Jung, 2007; Postnov et al., 2007). These models are of great potential interest for future modeling attempts but will not be considered here for simplicity.

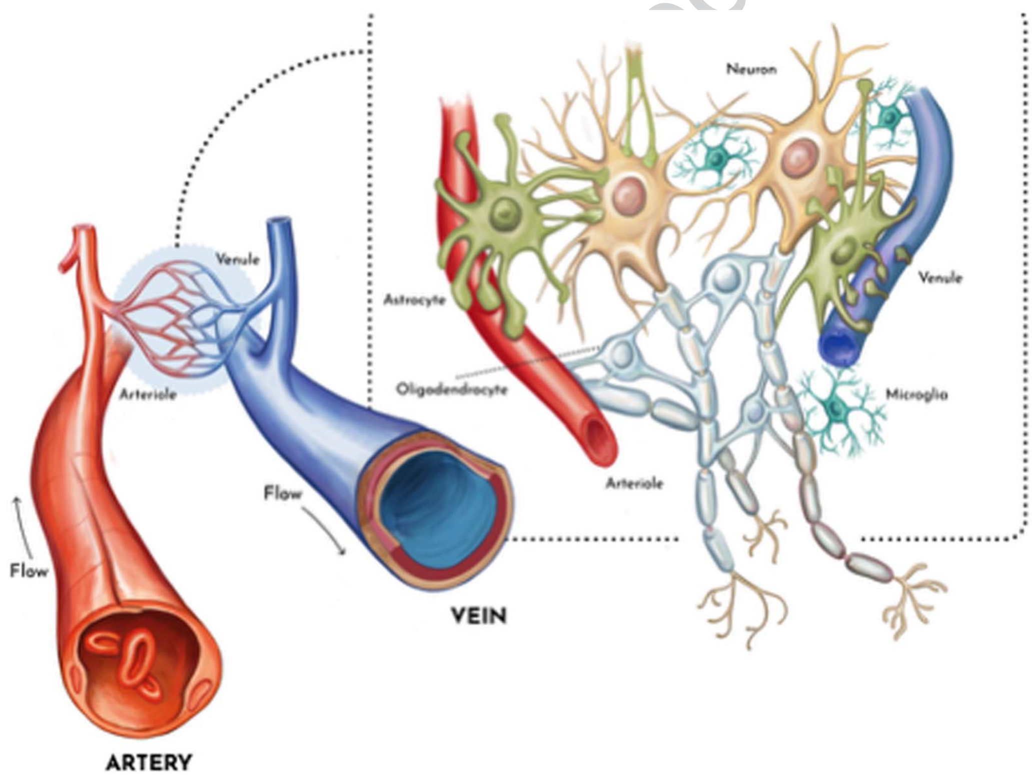

Figure 3. The neurogliovascular unit is here presented as the elementary unit for better modeling of brain morphology and function using generative computational models.

Now consider the stylized model introduced by Vértes (2023).

$$P_{i,j} = \left(d_{i,j}\right)^{\eta}\left(k_{i,j}\right)^{\gamma} \tag{1}$$

Where $P_{i,j}$ is the probability for a connection between two nodes i and j, $d_{i,j}$ is the wiring cost and $k_{i,j}$ is a bias term that promotes the clustering of regions with a large overlap in their existing connections. The exponents η and γ balance the two contributions d and k. These models have been able to reproduce the empirically observed network properties and distance distribution of human brain networks measured with neuroimaging but also from different modalities at different scales, in different species (Vértes, 2023). These stylized models have also been able to fit adequately the distribution of the same brain networks in psychiatric conditions and, specifically, of brains of patients with schizophrenia (Zhang et al., 2021).

In this form, these models do not inform on the underlying cellular machinery; however, they have the interesting property of allowing the linkage with biological microparameters of interest. We propose that, for example, the wiring cost parameter d can be explicitly factored by measured energetic quantities that can be experimentally measured, for example, resting potentials, action potentials in myelinated and nonmyelinated tracts, presynaptic and postsynaptic neurotransmitter activity, myelination by oligodendrocytes, glutamate/GABA recycling by astrocytes, and so forth (Attwell & Iadecola, 2002; Harris & Attwell, 2012), quantities that can be constrained by local metabolic capacity (Clarke & Sokoloff, 1999). Alterations of these parameters can be directly measured in vivo whenever technology allows or, alternatively, predicted by genetics. Individual polygenic risk scores have been shown to be associated with such parameterizations (Zhang et al., 2021); one step forward, albeit difficult, would be to predict alterations of metabolic costs due to genetic polymorphisms that induce loss or gain of function in the cellular processes described above (Sertbas & Ulgen, 2018). This formal scaling from genes-to-cells-to-systems is challenging albeit possible as we have recently demonstrated by using polygenic scores to predict volumetric changes across psychiatric disorders (Giacomel et al., 2026). Alternatively, these gene-to-function parameterization could be derived from induced pluripotent stem cells in appropriately designed genomic platforms (Brooks et al., 2022) (Figure 4).

CONCLUSION

In this manuscript, we have provided a critical view on how mathematical models that are used to simulate neural dynamic rely on a variety of properties that are directly tight with various aspects of brain biology. These models are bounded to provided simplified representations that may have difficulties in properly capturing the rich complexity of the cellular and molecular processes taking place underneath. Concretely, so far, they have mostly failed to include non-neuronal cells, glia, and the vascular system, which are important determinants of brain morphology and function. Importantly, most approaches miss the general principle that the brain's homeostasis is achieved by the set of interactions among cell types (e.g., the physiome [Hunter & Borg, 2003]), and that cell differentiation during development is itself a locally controlled process (Rowitch & Kriegstein, 2010).

The physiome:
The description of the functioning organism in normal and pathophysiological states. and is built upon the description of anatomical structure, chemical and biochemical composition, and material properties of an intact organism, including its genome, proteome, cell, tissue, and organ structures up to those of the whole intact being.

Given the lack of direct measurements in clinical populations, it is not surprising that the relevance of the neurogliovascular unit in neuropsychiatric disorders has not been matched by adequate computational modeling efforts. If direct measurements of these non-neuronal components and comprehensive molecular and cellular mapping will be made available, these components could allow us to further constraint the ranges of the model's parameters. This, in turn, has the potential to improve our ability to properly interpret the results obtained from

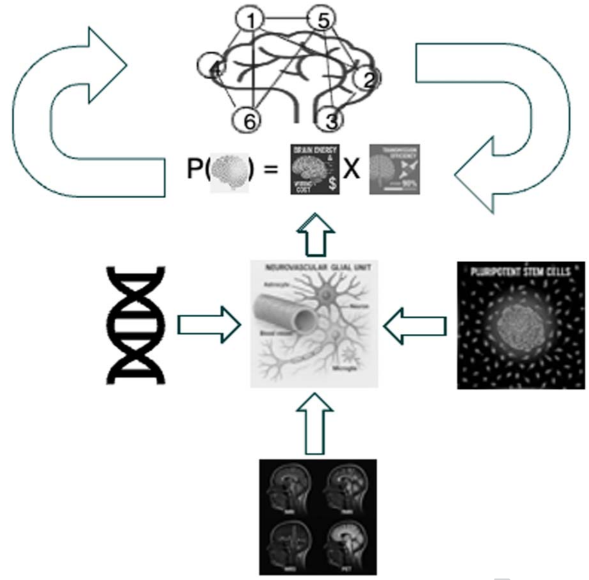

Figure 4. We propose that stylized models of brain network development can be used to test hypotheses on brain development and developmental disorders by incorporating genetic and functional information obtained from cellular models or in vivo imaging. This information can be inserted to evaluate the probability (P) of a connection as the results of the wiring and maintenance cost and efficiency in transmission due to clustering properties.

these models and contribute to the incremental, gradual process of discovery of the biology of brain in health and disease. This process is made more effective by the recent availability of ever more detailed and systematic information on the brain at any scale (e.g., molecular, cellular, and behavioral) (Calzavarini & Cevolani, 2022; Lawn et al., 2023; Markello et al., 2022; Yarkoni et al., 2011). For example, substantial improvement in model fit to functional MRI data has been observed by the inclusion of intracortical myelin density to constrain model by parameterizing synaptic strength (Deco et al., 2021; Demirtaş et al., 2019). Recently, the first molecular atlas of the cells that make up the human vasculature has been reported and has clearly resolved molecular components that are key to the pathophysiology of neurodegenerative disorders (Yang et al., 2022). Here we have proposed a general framework whereas cellular contribution from the neurogliovascular unit can be channeled in stylized models of network development.

All the above warrants great promise to the use of physical models and mathematical tools to study a wide range of topics with high potential translational value including, for example, the role of the vascular component and/or excitation-inhibition balance for several disorders affecting higher cognitive functions such as major depression, autism, and schizophrenia (Azmitia et al., 2016; Hanson & Gottesman, 2005; Hollestein et al., 2023; Liu et al., 2021). We hope this work may inspire more investigations on the rich interplay between neural, glial, and vascular activity, which we regard as critical to foster a greater understanding of the arguably most fascinating and mysterious organ, the brain.

ACKNOWLEDGMENTS

FET is partly funded by the NIHR Maudsley Biomedical Research Centre at South London and Maudsley NHS Foundation Trust and King's College London.

AUTHOR CONTRIBUTION

Federico E. Turkheimer: Conceptualization; Writing – original draft; Writing – review & editing. Fran Hancock: Conceptualization; Writing – original draft; Writing – review & editing. Fernando E. Rosas: Conceptualization; Writing – original draft; Writing – review & editing. Anthony C. Vernon: Conceptualization; Writing – original draft; Writing – review & editing. Deepak P. Srivastava: Conceptualization; Writing – original draft; Writing – review & editing. Po-Wah So: Conceptualization; Writing – original draft; Writing – review & editing. Andrea I. Luppi: Conceptualization; Writing – original draft; Writing – review & editing.

FUNDING INFORMATION

Federico Turkheimer, NIHR Maudsley Biomedical Research Centre (https://dx.doi.org/10.13039/100019418), Award ID: NIHR203318.

REFERENCES

Agnes, E. J., & Vogels, T. P. (2024). Co-dependent excitatory and inhibitory plasticity accounts for quick, stable and long-lasting memories in biological networks. *Nature Neuroscience*, *27*(5), 964–974. https://doi.org/10.1038/s41593-024-01597-4, PubMed: 38509348

Akarca, D., Vértes, P. E., Bullmore, E. T., CALM team., & Astle, D. E. (2021). A generative network model of neurodevelopmental diversity in structural brain organization. *Nature Communications*, *12*(1), 4216. https://doi.org/10.1038/s41467-021-24430-z, PubMed: 34244490

Araque, A., Parpura, V., Sanzgiri, R. P., & Haydon, P. G. (1998). Glutamate-dependent astrocyte modulation of synaptic transmission between cultured hippocampal neurons. *European Journal of Neuroscience*, *10*(6), 2129–2142. https://doi.org/10.1046/j.1460-9568.1998.00221.x, PubMed: 9753099

Arnatkeviciute, A., Fulcher, B. D., Oldham, S., Tiego, J., Paquola, C., Gerring, Z., ... Fornito, A. (2021). Genetic influences on hub connectivity of the human connectome. *Nature Communications*, *12*(1), 4237. https://doi.org/10.1038/s41467-021-24306-2, PubMed: 34244483

Attwell, D., & Iadecola, C. (2002). The neural basis of functional brain imaging signals. *Trends in Neurosciences*, *25*(12), 621–625. https://doi.org/10.1016/s0166-2236(02)02264-6, PubMed: 12446129

Azmitia, E. C., Saccomano, Z. T., Alzoobaee, M. F., Boldrini, M., & Whitaker-Azmitia, P. M. (2016). Persistent angiogenesis in the autism brain: An immunocytochemical study of postmortem cortex, brainstem and cerebellum. *Journal of Autism and Developmental Disorders*, *46*(4), 1307–1318. https://doi.org/10.1007/s10803-015-2672-6, PubMed: 26667147

Balchandani, P., & Naidich, T. P. (2015). Ultra-high-field MR neuroimaging. *American Journal of Neuroradiology*, *36*(7), 1204–1215. https://doi.org/10.3174/ajnr.A4180, PubMed: 25523591

Barabási, D. L., Bianconi, G., Bullmore, E., Burgess, M., Chung, S., Eliassi-Rad, T., ... Buzsáki, G. (2023). Neuroscience needs network science. *Journal of Neuroscience*, *43*(34), 5989–5995. https://doi.org/10.1523/JNEUROSCI.1014-23.2023, PubMed: 37612141

Bassett, D. S., & Bullmore, E. T. (2017). Small-world brain networks revisited. *Neuroscientist*, *23*(5), 499–516. https://doi.org/10.1177/1073858416667720, PubMed: 27655008

Bassett, D. S., Meyer-Lindenberg, A., Achard, S., Duke, T., & Bullmore, E. (2006). Adaptive reconfiguration of fractal small-world human brain functional networks. *Proceedings of the National Academy of Sciences of the United States of America*, *103*(51), 19518–19523. https://doi.org/10.1073/pnas.0606005103, PubMed: 17159150

Bassett, D. S., Zurn, P., & Gold, J. I. (2018). On the nature and use of models in network neuroscience. *Nature Reviews of Neuroscience*, *19*(9), 566–578. https://doi.org/10.1038/s41583-018-0038-8, PubMed: 30002509

Bassingthwaighte, J. B. (1992). Fractal vascular growth patterns. *Acta Stereologica*, *11*(Suppl. 1), 305–319. PubMed: 25237211

Beaulieu, C. (1993). Numerical data on neocortical neurons in adult rat, with special reference to the GABA population. *Brain Research*, *609*(1–2), 284–292. https://doi.org/10.1016/0006-8993(93)90884-p, PubMed: 8508310

Belan, P. V., & Kostyuk, P. G. (2002). Glutamate-receptor-induced modulation of GABAergic synaptic transmission in the hippocampus. *Pflügers Archiv*, *444*(1–2), 26–37. https://doi.org/10.1007/s00424-002-0783-3, PubMed: 11976913

Beul, S. F., Goulas, A., & Hilgetag, C. C. (2018). Comprehensive computational modelling of the development of mammalian cortical connectivity underlying an architectonic type principle. *PLoS Computational Biology*, *14*(11), e1006550. https://doi.org/10.1371/journal.pcbi.1006550, PubMed: 30475798

Börgers, C., & Kopell, N. (2003). Synchronization in networks of excitatory and inhibitory neurons with sparse, random connectivity. *Neural Computation*, *15*(3), 509–538. https://doi.org/10.1162/089976603321192059, PubMed: 12620157

Brambilla, P., Cerini, R., Fabene, P. F., Andreone, N., Rambaldelli, G., Farace, P., ... Tansella, M. (2007). Assessment of cerebral blood volume in schizophrenia: A magnetic resonance imaging study. *Journal of Psychiatric Research*, *41*(6), 502–510. https://doi.org/10.1016/j.jpsychires.2006.03.002, PubMed: 16698038

Breakspear, M. (2017). Dynamic models of large-scale brain activity. *Nature Neuroscience*, *20*(3), 340–352. https://doi.org/10.1038/nn.4497, PubMed: 28230845

Brembs, B. (2021). The brain as a dynamically active organ. *Biochemical and Biophysical Research Communications*, *564*, 55–69. https://doi.org/10.1016/j.bbrc.2020.12.011, PubMed: 33317833

Bremermann, H. J. (1963). Limits of genetic control. *IEEE Transactions on Military Electronics*, *MIL-7*(2/3), 200–205. https://doi.org/10.1109/TME.1963.4323073

Brooks, I. R., Garrone, C. M., Kerins, C., Kiar, C. S., Syntaka, S., Xu, J. Z., ... Watt, F. M. (2022). Functional genomics and the future of iPSCs in disease modeling. *Stem Cell Reports*, *17*(5), 1033–1047. https://doi.org/10.1016/j.stemcr.2022.03.019, PubMed: 35487213

Brynildsen, J. K., Rajan, K., Henderson, M. X., & Bassett, D. S. (2023). Network models to enhance the translational impact of cross-species studies. *Nature Reviews Neuroscience*, *24*(9), 575–588. https://doi.org/10.1038/s41583-023-00720-x, PubMed: 37524935

Bullmore, E., & Sporns, O. (2012). The economy of brain network organization. *Nature Reviews Neuroscience*, *13*(5), 336–349. https://doi.org/10.1038/nrn3214, PubMed: 22498897

Buzsáki, G., & Wang, X.-J. (2012). Mechanisms of gamma oscillations. *Annual Review of Neuroscience*, *35*, 203–225. https://doi.org/10.1146/annurev-neuro-062111-150444, PubMed: 22443509

Cabral, J., Hugues, E., Sporns, O., & Deco, G. (2011). Role of local network oscillations in resting-state functional connectivity. *NeuroImage*, *57*(1), 130–139. https://doi.org/10.1016/j.neuroimage.2011.04.010, PubMed: 21511044

Calzavarini, F., & Cevolani, G. (2022). Abductive reasoning in cognitive neuroscience: Weak and strong reverse inference. *Synthese*, *200*(2), 70. https://doi.org/10.1007/s11229-022-03585-2

Captur, G., Karperien, A. L., Hughes, A. D., Francis, D. P., & Moon, J. C. (2017). The fractal heart - embracing mathematics in the cardiology clinic. *Nature Reviews Cardiology*, *14*(1), 56–64. https://doi.org/10.1038/nrcardio.2016.161, PubMed: 27708281

Cardy, J., & Täuber, U. C. (1996). Theory of branching and annihilating random walks. *Physical Review Letters*, *77*(23), 4780–4783. https://doi.org/10.1103/PhysRevLett.77.4780, PubMed: 10062629

Carmeliet, P., & Tessier-Lavigne, M. (2005). Common mechanisms of nerve and blood vessel wiring. *Nature*, *436*(7048), 193–200. https://doi.org/10.1038/nature03875, PubMed: 16015319

Carrier, M., Guilbert, J., Lévesque, J.-P., Tremblay, M.-E., & Desjardins, M. (2020). Structural and functional features of developing brain capillaries, and their alteration in schizophrenia. *Frontiers in Cellular Neuroscience*, *14*, 595002. https://doi.org/10.3389/fncel.2020.595002, PubMed: 33519380

Casas, B. S., Vitória, G., do Costa, M. N., Madeiro da Costa, R., Trindade, P., Maciel, R., ... Palma, V. (2018). hiPSC-derived neural stem cells from patients with schizophrenia induce an impaired angiogenesis. *Translational Psychiatry*, *8*(1), 48. https://doi.org/10.1038/s41398-018-0095-9, PubMed: 29467462

Caserta, F., Eldred, W. D., Fernandez, E., Hausman, R. E., Stanford, L. R., Bulderev, S. V., ... Stanley, H. E. (1995). Determination of fractal dimension of physiologically characterized neurons in two and three dimensions. *Journal of Neuroscience Methods*, *56*(2), 133–144. https://doi.org/10.1016/0165-0270(94)00115-w, PubMed: 7752679

Castaldo, F., Pascoa Dos Santos, F., Timms, R. C., Cabral, J., Vohryzek, J., Deco, G., ... Litvak, V. (2023). Multi-modal and multi-model interrogation of large-scale functional brain networks. *NeuroImage*, *277*, 120236. https://doi.org/10.1016/j.neuroimage.2023.120236, PubMed: 37355200

Clarke, D. D., & Sokoloff, L. (1999). Regulation of cerebral metabolic rate. In G. J. Siegel, B. W. Agranoff, & R. W. Albers (Eds.), *Basic neurochemistry: Molecular, cellular and medical aspects*. Lippincott-Raven.

Clark-Raymond, A., & Halaris, A. (2013). VEGF and depression: A comprehensive assessment of clinical data. *Journal of Psychiatric Research*, *47*(8), 1080–1087. https://doi.org/10.1016/j.jpsychires.2013.04.008, PubMed: 23684549

Collignon, A., Dion-Albert, L., Ménard, C., & Coelho-Santos, V. (2024). Sex, hormones and cerebrovascular function: From development to disorder. *Fluids and Barriers of the CNS*, *21*(1), 2. https://doi.org/10.1186/s12987-023-00496-3, PubMed: 38178239

Colombo, M. A., Napolitani, M., Boly, M., Gosseries, O., Casarotto, S., Rosanova, M., ... Sarasso, S. (2019). The spectral exponent of the resting EEG indexes the presence of consciousness during unresponsiveness induced by propofol, xenon, and ketamine. *NeuroImage*, *189*, 631–644. https://doi.org/10.1016/j.neuroimage.2019.01.024, PubMed: 30639334

D'amour, J. A., & Froemke, R. C. (2015). Inhibitory and excitatory spike-timing-dependent plasticity in the auditory cortex. *Neuron*, *86*(2), 514–528. https://doi.org/10.1016/j.neuron.2015.03.014, PubMed: 25843405

Deco, G., Jirsa, V., McIntosh, A. R., Sporns, O., & Kötter, R. (2009). Key role of coupling, delay, and noise in resting brain fluctuations. *Proceedings of the National Academy of Sciences of the United States of America*, *106*(25), 10302–10307. https://doi.org/10.1073/pnas.0901831106, PubMed: 19497858

Deco, G., Kringelbach, M. L., Arnatkeviciute, A., Oldham, S., Sabaroedin, K., Rogasch, N. C., ... Fornito, A. (2021). Dynamical consequences of regional heterogeneity in the brain's transcriptional landscape. *Science Advances*, *7*(29), eabf4752. https://doi.org/10.1126/sciadv.abf4752, PubMed: 34261652

Deco, G., Ponce-Alvarez, A., Hagmann, P., Romani, G. L., Mantini, D., & Corbetta, M. (2014). How local excitation-inhibition ratio impacts the whole brain dynamics. *Journal of Neuroscience*, *34*(23), 7886–7898. https://doi.org/10.1523/jneurosci.5068-13.2014, PubMed: 24899711

Demirtaş, M., Burt, J. B., Helmer, M., Ji, J. L., Adkinson, B. D., Glasser, M. F., ... Murray, J. D. (2019). Hierarchical heterogeneity across human cortex shapes large-scale neural dynamics. *Neuron*, *101*(6), 1181–1194. https://doi.org/10.1016/j.neuron.2019.01.017, PubMed: 30744986

Duclos, K. K., Hendrikse, J. L., & Jamniczky, H. A. (2019). Investigating the evolution and development of biological complexity under the framework of epigenetics. *Evolution & Development*, *21*(5), 247–264. https://doi.org/10.1111/ede.12301, PubMed: 31268245

Eichmann, A., & Thomas, J.-L. (2013). Molecular parallels between neural and vascular development. *Cold Spring Harbor Perspectives in Medicine*, *3*(1), a006551. https://doi.org/10.1101/cshperspect.a006551, PubMed: 23024177

Fields, R. D., Araque, A., Johansen-Berg, H., Lim, S.-S., Lynch, G., Nave, K.-A., ... Wake, H. (2014). Glial biology in learning and cognition. *Neuroscientist*, *20*(5), 426–431. https://doi.org/10.1177/1073858413504465, PubMed: 24122821

Flanagan, B., McDaid, L., Wade, J. J., Toman, M., Wong-Lin, K., & Harkin, J. (2021). A computational study of astrocytic GABA release at the glutamatergic synapse: EAAT-2 and GAT-3 coupled dynamics. *Frontiers in Cellular Neuroscience*, *15*, 682460. https://doi.org/10.3389/fncel.2021.682460, PubMed: 34322000

Fornito, A., Zalesky, A., & Breakspear, M. (2015). The connectomics of brain disorders. *Nature Reviews Neuroscience*, *16*(3), 159–172. https://doi.org/10.1038/nrn3901, PubMed: 25697159

Friston, K. J. (1997). Transients, metastability, and neuronal dynamics. *NeuroImage*, *5*(2), 164–171. https://doi.org/10.1006/nimg.1997.0259, PubMed: 9345546

Gao, R., Peterson, E. J., & Voytek, B. (2017). Inferring synaptic excitation/inhibition balance from field potentials. *NeuroImage*, *158*, 70–78. https://doi.org/10.1016/j.neuroimage.2017.06.078, PubMed: 28676297

Gerhardt, H., Golding, M., Fruttiger, M., Ruhrberg, C., Lundkvist, A., Abramsson, A., ... Betsholtz, C. (2003). VEGF guides angiogenic sprouting utilizing endothelial tip cell filopodia. *Journal of Cell Biology*, *161*(6), 1163–1177. https://doi.org/10.1083/jcb.200302047, PubMed: 12810700

Giacomel, A., Powell, T. R., Duarte, R. R. R., Nordio, G., Williams, S. C. R., Turkheimer, F., ... Dima, D. (2026). Transcriptome-informed brain cartography of polygenic risk and association with brain structure in major psychiatric disorders. *Molecular Psychiatry*. https://doi.org/10.1038/s41380-026-03497-4, PubMed: 41792457

Gollo, L. L., & Breakspear, M. (2014). The frustrated brain: From dynamics on motifs to communities and networks. *Philosophical Transactions of the Royal Society of London B: Biological Sciences*, *369*(1653), 20130532. https://doi.org/10.1098/rstb.2013.0532, PubMed: 25180310

Goulas, A., Changeux, J.-P., Wagstyl, K., Amunts, K., Palomero-Gallagher, N., & Hilgetag, C. C. (2021). The natural axis of transmitter receptor distribution in the human cerebral cortex. *Proceedings of the National Academy of Sciences of the United States of America*, *118*(3), e2020574118. https://doi.org/10.1073/pnas.2020574118, PubMed: 33452137

Guidolin, D., Fede, C., & Tortorella, C. (2019). Nerve cells developmental processes and the dynamic role of cytokine signaling. *International Journal of Developmental Neuroscience*, *77*, 3–17. https://doi.org/10.1016/j.ijdevneu.2018.11.003, PubMed: 30465872

Han, J., Kesner, P., Metna-Laurent, M., Duan, T., Xu, L., Georges, F., ... Zhang, X. (2012). Acute cannabinoids impair working memory through astroglial CB1 receptor modulation of hippocampal LTD. *Cell*, *148*(5), 1039–1050. https://doi.org/10.1016/j.cell.2012.01.037, PubMed: 22385967

Hancock, F., Rosas, F. E., Luppi, A. I., Zhang, M., Mediano, P. A. M., Cabral, J., ... Turkheimer, F. E. (2025). Metastability demystified - the foundational past, the pragmatic present and the promising future. *Nature Reviews Neuroscience*, *26*(2), 82–100. https://doi.org/10.1038/s41583-024-00883-1, PubMed: 39663408

Hancock, F., Rosas, F. E., Mediano, P. A. M., Luppi, A. I., Cabral, J., Dipasquale, O., & Turkheimer, F. E. (2022). May the 4C's be with you: An overview of complexity-inspired frameworks for analysing resting-state neuroimaging data. *Journal of the Royal Society Interface*, *19*(191), 20220214. https://doi.org/10.1098/rsif.2022.0214, PubMed: 35765805

Hannezo, E., Scheele, C. L. G. J., Moad, M., Drogo, N., Heer, R., Sampogna, R. V., ... Simons, B. D. (2017). A unifying theory of branching morphogenesis. *Cell*, *171*(1), 242–255. https://doi.org/10.1016/j.cell.2017.08.026, PubMed: 28938116

Hanson, D. R., & Gottesman, I. I. (2005). Theories of schizophrenia: A genetic-inflammatory-vascular synthesis. *BMC Medical Genetics*, *6*, 7. https://doi.org/10.1186/1471-2350-6-7, PubMed: 15707482

Harris, J. J., & Attwell, D. (2012). The energetics of CNS white matter. *Journal of Neuroscience*, *32*(1), 356–371. https://doi.org/10.1523/JNEUROSCI.3430-11.2012, PubMed: 22219296

Hellyer, P. J., Scott, G., Shanahan, M., Sharp, D. J., & Leech, R. (2015). Cognitive flexibility through metastable neural dynamics is disrupted by damage to the structural connectome. *Journal of Neuroscience*, *35*(24), 9050–9063. https://doi.org/10.1523/JNEUROSCI.4648-14.2015, PubMed: 26085630

Helmstaedter, M., Sakmann, B., & Feldmeyer, D. (2009). Neuronal correlates of local, lateral, and translaminar inhibition with reference to cortical columns. *Cerebral Cortex*, *19*(4), 926–937. https://doi.org/10.1093/cercor/bhn141, PubMed: 18832335

Herculano-Houzel, S., & Rothman, D. L. (2022). From a demand-based to a supply-limited framework of brain metabolism. *Frontiers in Integrative Neuroscience*, *16*, 818685. https://doi.org/10.3389/fnint.2022.818685, PubMed: 35431822

Hevner, R. F., Daza, R. A. M., Englund, C., Kohtz, J., & Fink, A. (2004). Postnatal shifts of interneuron position in the neocortex of normal and reeler mice: Evidence for inward radial migration. *Neuroscience*, *124*(3), 605–618. https://doi.org/10.1016/j.neuroscience.2003.11.033, PubMed: 14980731

Hodgkin, A. L., & Huxley, A. F. (1990). A quantitative description of membrane current and its application to conduction and excitation in nerve. *Bulletin of Mathematical Biology*, *52*(1–2), 25–71. https://doi.org/10.1007/BF02459568, PubMed: 2185861

Hollestein, V., Poelmans, G., Forde, N. J., Beckmann, C. F., Ecker, C., Mann, C., ... Naaijen, J. (2023). Excitatory/inhibitory imbalance in autism: The role of glutamate and GABA gene-sets in symptoms and cortical brain structure. *Translational Psychiatry*, *13*(1), 18. https://doi.org/10.1038/s41398-023-02317-5, PubMed: 36681677

Honey, C. J., Sporns, O., Cammoun, L., Gigandet, X., Thiran, J. P., Meuli, R., & Hagmann, P. (2009). Predicting human resting-state functional connectivity from structural connectivity. *Proceedings of the National Academy of Sciences of the United States of America*, *106*(6), 2035–2040. https://doi.org/10.1073/pnas.0811168106, PubMed: 19188601

Hoseth, E. Z., Krull, F., Dieset, I., Mørch, R. H., Hope, S., Gardsjord, E. S., ... Ueland, T. (2018). Attenuated Notch signaling in

schizophrenia and bipolar disorder. *Scientific Reports*, *8*(1), 5349. https://doi.org/10.1038/s41598-018-23703-w, PubMed: 29593239
Howarth, C., Gleeson, P., & Attwell, D. (2012). Updated energy budgets for neural computation in the neocortex and cerebellum. *Journal of Cerebral Blood Flow and Metabolism*, *32*(7), 1222–1232. https://doi.org/10.1038/jcbfm.2012.35, PubMed: 22434069
Howarth, C., Peppiatt-Wildman, C. M., & Attwell, D. (2010). The energy use associated with neural computation in the cerebellum. *Journal of Cerebral Blood Flow and Metabolism*, *30*(2), 403–414. https://doi.org/10.1038/jcbfm.2009.231, PubMed: 19888288
Hua, J., Brandt, A. S., Lee, S., Blair, N. I. S., Wu, Y., Lui, S., … Margolis, R. L. (2017). Abnormal grey matter arteriolar cerebral blood volume in schizophrenia measured with 3D inflow-based vascular-space-occupancy MRI at 7T. *Schizophrenia Bulletin*, *43*(3), 620–632. https://doi.org/10.1093/schbul/sbw109, PubMed: 27539951
Hua, J., Qin, Q., Donahue, M. J., Zhou, J., Pekar, J. J., & van Zijl, P. C. M. (2011). Inflow-based vascular-space-occupancy (iVASO) MRI. *Magnetic Resonance in Medicine*, *66*(1), 40–56. https://doi.org/10.1002/mrm.22775, PubMed: 21695719
Hunter, P. J., & Borg, T. K. (2003). Integration from proteins to organs: The Physiome Project. *Nature Reviews Molecular Cell Biology*, *4*(3), 237–243. https://doi.org/10.1038/nrm1054, PubMed: 12612642
Iadecola, C. (2017). The neurovascular unit coming of age: A journey through neurovascular coupling in health and disease. *Neuron*, *96*(1), 17–42. https://doi.org/10.1016/j.neuron.2017.07.030, PubMed: 28957666
Ii, S., Kitade, H., Ishida, S., Imai, Y., Watanabe, Y., & Wada, S. (2020). Multiscale modeling of human cerebrovasculature: A hybrid approach using image-based geometry and a mathematical algorithm. *PLoS Computational Biology*, *16*(6), e1007943. https://doi.org/10.1371/journal.pcbi.1007943, PubMed: 32569287
Isaacson, J. S., & Scanziani, M. (2011). How inhibition shapes cortical activity. *Neuron*, *72*(2), 231–243. https://doi.org/10.1016/j.neuron.2011.09.027, PubMed: 22017986
Izhikevich, E. M., & Edelman, G. M. (2008). Large-scale model of mammalian thalamocortical systems. *Proceedings of the National Academy of Sciences of the United States of America*, *105*(9), 3593–3598. https://doi.org/10.1073/pnas.0712231105, PubMed: 18292226
Jelen, L. A., King, S., Mullins, P. G., & Stone, J. M. (2018). Beyond static measures: A review of functional magnetic resonance spectroscopy and its potential to investigate dynamic glutamatergic abnormalities in schizophrenia. *Journal of Psychopharmacology*, *32*(5), 497–508. https://doi.org/10.1177/0269881117747579, PubMed: 29368979
Jones, C. A., & Li, D. Y. (2007). Common cues regulate neural and vascular patterning. *Current Opinion in Genetics & Development*, *17*(4), 332–336. https://doi.org/10.1016/j.gde.2007.07.004, PubMed: 17692512
Jourdain, P., Bergersen, L. H., Bhaukaurally, K., Bezzi, P., Santello, M., Domercq, M., … Volterra, A. (2007). Glutamate exocytosis from astrocytes controls synaptic strength. *Nature Neuroscience*, *10*(3), 331–339. https://doi.org/10.1038/nn1849, PubMed: 17310248
Kalil, K., & Dent, E. W. (2014). Branch management: Mechanisms of axon branching in the developing vertebrate CNS. *Nature Reviews Neuroscience*, *15*(1), 7–18. https://doi.org/10.1038/nrn3650, PubMed: 24356070
Kang, J., Jiang, L., Goldman, S. A., & Nedergaard, M. (1998). Astrocyte-mediated potentiation of inhibitory synaptic transmission. *Nature Neuroscience*, *1*(8), 683–692. https://doi.org/10.1038/3684, PubMed: 10196584
Karch, R., Neumann, F., Neumann, M., & Schreiner, W. (1999). A three-dimensional model for arterial tree representation, generated by constrained constructive optimization. *Computers in Biology and Medicine*, *29*(1), 19–38. https://doi.org/10.1016/s0010-4825(98)00045-6, PubMed: 10207653
Katsel, P., Roussos, P., Pletnikov, M., & Haroutunian, V. (2017). Microvascular anomaly conditions in psychiatric disease. Schizophrenia - angiogenesis connection. *Neuroscience and Biobehavioral Reviews*, *77*, 327–339. https://doi.org/10.1016/j.neubiorev.2017.04.003, PubMed: 28396239
Kauffman, S. A. (1991). Antichaos and adaptation. *Scientific American*, *265*(2), 78–84. https://doi.org/10.1038/scientificamerican0891-78, PubMed: 1862333
Kersanté, F., Rowley, S. C. S., Pavlov, I., Gutièrrez-Mecinas, M., Semyanov, A., Reul, J. M. H. M., … Linthorst, A. C. E. (2013). A functional role for both-aminobutyric acid (GABA) transporter-1 and GABA transporter-3 in the modulation of extracellular GABA and GABAergic tonic conductances in the rat hippocampus. *Journal of Physiology*, *591*(10), 2429–2441. https://doi.org/10.1113/jphysiol.2012.246298, PubMed: 23381899
Kerstjens, S., Michel, G., & Douglas, R. J. (2022). Constructive connectomics: How neuronal axons get from here to there using gene-expression maps derived from their family trees. *PLoS Computational Biology*, *18*(8), e1010382. https://doi.org/10.1371/journal.pcbi.1010382, PubMed: 36006873
Kim, J., Kwon, N., Chang, S., Kim, K.-T., Lee, D., Kim, S., … Rhyu, I. J. (2011). Altered branching patterns of Purkinje cells in mouse model for cortical development disorder. *Scientific Reports*, *1*, 122. https://doi.org/10.1038/srep00122, PubMed: 22355639
Kirchner, J. H., & Gjorgjieva, J. (2021). Emergence of local and global synaptic organization on cortical dendrites. *Nature Communications*, *12*(1), 4005. https://doi.org/10.1038/s41467-021-23557-3, PubMed: 34183661
Kong, X., Kong, R., Orban, C., Wang, P., Zhang, S., Anderson, K., … Yeo, B. T. T. (2021). Sensory-motor cortices shape functional connectivity dynamics in the human brain. *Nature Communications*, *12*(1), 6373. https://doi.org/10.1038/s41467-021-26704-y, PubMed: 34737302
Kumar, B. S., Menon, S. C., Gayathri, S. R., & Chakravarthy, V. S. (2022). The influence of neural activity and neural cytoarchitecture on cerebrovascular arborization: A computational model. *Frontiers in Neuroscience*, *16*, 917196. https://doi.org/10.3389/fnins.2022.917196, PubMed: 35860300
Lawn, T., Howard, M. A., Turkheimer, F., Misic, B., Deco, G., Martins, D., & Dipasquale, O. (2023). From neurotransmitters to networks: Transcending organisational hierarchies with

molecular-informed functional imaging. *Neuroscience and Biobehavioral Reviews*, *150*, 105193. https://doi.org/10.1016/j.neubiorev.2023.105193, PubMed: 37086932

Lea-Carnall, C. A., El-Deredy, W., Stagg, C. J., Williams, S. R., & Trujillo-Barreto, N. J. (2023). A mean-field model of glutamate and GABA synaptic dynamics for functional MRS. *NeuroImage*, *266*, 119813. https://doi.org/10.1016/j.neuroimage.2022.119813, PubMed: 36528313

Lee, B.-H., & Kim, Y.-K. (2012). Increased plasma VEGF levels in major depressive or manic episodes in patients with mood disorders. *Journal of Affective Disorders*, *136*(1–2), 181–184. https://doi.org/10.1016/j.jad.2011.07.021, PubMed: 21862441

Libby, J., Marghoub, A., Johnson, D., Khonsari, R. H., Fagan, M. J., & Moazen, M. (2017). Modelling human skull growth: A validated computational model. *Journal of Royal Society Interface*, *14*(130), 20170202. https://doi.org/10.1098/rsif.2017.0202, PubMed: 28566514

Lim, L., Mi, D., Llorca, A., & Marín, O. (2018). Development and functional diversification of cortical interneurons. *Neuron*, *100*(2), 294–313. https://doi.org/10.1016/j.neuron.2018.10.009, PubMed: 30359598

Lindenmayer, A. (1968). Mathematical models for cellular interactions in development. I. Filaments with one-sided inputs. *Journal of Theoretical Biology*, *18*(3), 280–299. https://doi.org/10.1016/0022-5193(68)90079-9, PubMed: 5659071

Litwin-Kumar, A., & Doiron, B. (2014). Formation and maintenance of neuronal assemblies through synaptic plasticity. *Nature Communications*, *5*, 5319. https://doi.org/10.1038/ncomms6319, PubMed: 25395015

Liu, Q., Xu, Q., Arcuino, G., Kang, J., & Nedergaard, M. (2004). Astrocyte-mediated activation of neuronal kainate receptors. *Proceedings of the National Academy of Sciences of the United States of America*, *101*(9), 3172–3177. https://doi.org/10.1073/pnas.0306731101, PubMed: 14766987

Liu, Y., Liang, M., Zhou, Y., He, Y., Hao, Y., Song, M., ... Jiang, T. (2008). Disrupted small-world networks in schizophrenia. *Brain*, *131*(Pt 4), 945–961. https://doi.org/10.1093/brain/awn018, PubMed: 18299296

Liu, Y., Ouyang, P., Zheng, Y., Mi, L., Zhao, J., Ning, Y., & Guo, W. (2021). A selective review of the excitatory-inhibitory imbalance in schizophrenia: Underlying biology, genetics, microcircuits, and symptoms. *Frontiers in Cell and Developmental Biology*, *9*, 664535. https://doi.org/10.3389/fcell.2021.664535, PubMed: 34746116

Lodato, S., Rouaux, C., Quast, K. B., Jantrachotechatchawan, C., Studer, M., Hensch, T. K., & Arlotta, P. (2011). Excitatory projection neuron subtypes control the distribution of local inhibitory interneurons in the cerebral cortex. *Neuron*, *69*(4), 763–779. https://doi.org/10.1016/j.neuron.2011.01.015, PubMed: 21338885

Lord, L.-D., Expert, P., Huckins, J. F., & Turkheimer, F. E. (2013). Cerebral energy metabolism and the brain's functional network architecture: An integrative review. *Journal of Cerebral Blood Flow and Metabolism*, *33*(9), 1347–1354. https://doi.org/10.1038/jcbfm.2013.94, PubMed: 23756687

Lord, L.-D., Stevner, A. B., Deco, G., & Kringelbach, M. L. (2017). Understanding principles of integration and segregation using whole-brain computational connectomics: Implications for neuropsychiatric disorders. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, *375*(2096), 20160283. https://doi.org/10.1098/rsta.2016.0283, PubMed: 28507228

Luppi, A. I., Rosas, F. E., Mediano, P. A. M., Menon, D. K., & Stamatakis, E. A. (2024). Information decomposition and the informational architecture of the brain. *Trends in Cognitive Sciences*, *28*(4), 352–368. https://doi.org/10.1016/j.tics.2023.11.005, PubMed: 38199949

Luppi, A. I., Rosas, F. E., Noonan, M. P., Mediano, P. A. M., Kringelbach, M. L., Carhart-Harris, R. L., ... Turkheimer, F. E. (2024). Oxygen and the spark of human brain evolution: Complex interactions of metabolism and cortical expansion across development and evolution. *Neuroscientist*, *30*(2), 173–198. https://doi.org/10.1177/10738584221138032, PubMed: 36476177

Mapelli, J., Gandolfi, D., Vilella, A., Zoli, M., & Bigiani, A. (2016). Heterosynaptic GABAergic plasticity bidirectionally driven by the activity of pre- and postsynaptic NMDA receptors. *Proceedings of the National Academy of Sciences of the United States of America*, *113*(35), 9898–9903. https://doi.org/10.1073/pnas.1601194113, PubMed: 27531957

Marcaggi, P., & Attwell, D. (2004). Role of glial amino acid transporters in synaptic transmission and brain energetics. *Glia*, *47*(3), 217–225. https://doi.org/10.1002/glia.20027, PubMed: 15252810

Markello, R. D., Hansen, J. Y., Liu, Z.-Q., Bazinet, V., Shafiei, G., Suárez, L. E., ... Misic, B. (2022). Neuromaps: Structural and functional interpretation of brain maps. *Nature Methods*, *19*(11), 1472–1479. https://doi.org/10.1038/s41592-022-01625-w, PubMed: 36203018

Meier, M. H., Shalev, I., Moffitt, T. E., Kapur, S., Keefe, R. S. E., Wong, T. Y., ... Poulton, R. (2013). Microvascular abnormality in schizophrenia as shown by retinal imaging. *American Journal of Psychiatry*, *170*(12), 1451–1459. https://doi.org/10.1176/appi.ajp.2013.13020234, PubMed: 24030514

Menassa, D. A., & Gomez-Nicola, D. (2018). Microglial dynamics during human brain development. *Frontiers in Immunology*, *9*, 1014. https://doi.org/10.3389/fimmu.2018.01014, PubMed: 29881376

Meyer, H. S., Schwarz, D., Wimmer, V. C., Schmitt, A. C., Kerr, J. N. D., Sakmann, B., & Helmstaedter, M. (2011). Inhibitory interneurons in a cortical column form hot zones of inhibition in layers 2 and 5A. *Proceedings of the National Academy of Sciences of the United States of America*, *108*(40), 16807–16812. https://doi.org/10.1073/pnas.1113648108, PubMed: 21949377

Milosević, N. T., & Ristanović, D. (2006). Fractality of dendritic arborization of spinal cord neurons. *Neuroscience Letters*, *396*(3), 172–176. https://doi.org/10.1016/j.neulet.2005.11.031, PubMed: 16364544

Min, R., & Nevian, T. (2012). Astrocyte signaling controls spike timing-dependent depression at neocortical synapses. *Nature Neuroscience*, *15*(5), 746–753. https://doi.org/10.1038/nn.3075, PubMed: 22446881

Miyoshi, G., Hjerling-Leffler, J., Karayannis, T., Sousa, V. H., Butt, S. J. B., Battiste, J., ... Fishell, G. (2010). Genetic fate mapping reveals that the caudal ganglionic eminence produces a large and diverse population of superficial cortical interneurons. *Journal of Neuroscience*, *30*(5), 1582–1594. https://doi.org/10.1523/JNEUROSCI.4515-09.2010, PubMed: 20130169

Morigiwa, K., Tauchi, M., & Fukuda, Y. (1989). Fractal analysis of ganglion cell dendritic branching patterns of the rat and cat retinae. *Neuroscience Research Supplement*, *10*, S131–S139. https://doi.org/10.1016/0921-8696(89)90015-7, PubMed: 2594252

Muldoon, S. F., Bridgeford, E. W., & Bassett, D. S. (2016). Small-world propensity and weighted brain networks. *Scientific Reports*, *6*(1), 22057. https://doi.org/10.1038/srep22057, PubMed: 26912196

Nadkarni, S., & Jung, P. (2007). Modeling synaptic transmission of the tripartite synapse. *Physical Biology*, *4*(1), 1–9. https://doi.org/10.1088/1478-3975/4/1/001, PubMed: 17406080

Naskar, A., Vattikonda, A., Deco, G., Roy, D., & Banerjee, A. (2021). Multiscale dynamic mean field (MDMF) model relates resting-state brain dynamics with local cortical excitatory-inhibitory neurotransmitter homeostasis. *Network Neuroscience*, *5*(3), 757–782. https://doi.org/10.1162/netn_a_00197, PubMed: 34746626

Navarrete, M., Perea, G., Fernandez de Sevilla, D., Gómez-Gonzalo, M., Núñez, A., Martín, E. D., & Araque, A. (2012). Astrocytes mediate in vivo cholinergic-induced synaptic plasticity. *PLoS Biology*, *10*(2), e1001259. https://doi.org/10.1371/journal.pbio.1001259, PubMed: 22347811

Nunes, F. D. D., Ferezin, L. P., Pereira, S. C., Figaro-Drumond, F. V., Pinheiro, L. C., Menezes, I. C., … Lacchini, R. (2022). The association of biochemical and genetic biomarkers in VEGF pathway with depression. *Pharmaceutics*, *14*(12), 2757. https://doi.org/10.3390/pharmaceutics14122757, PubMed: 36559251

Ofer, N., Shefi, O., & Yaari, G. (2017). Branching morphology determines signal propagation dynamics in neurons. *Scientific Reports*, *7*(1), 8877. https://doi.org/10.1038/s41598-017-09184-3, PubMed: 28827727

Okun, M., & Lampl, I. (2008). Instantaneous correlation of excitation and inhibition during ongoing and sensory-evoked activities. *Nature Neuroscience*, *11*(5), 535–537. https://doi.org/10.1038/nn.2105, PubMed: 18376400

Oldham, S., Fulcher, B. D., Aquino, K., Arnatkevičiūtė, A., Paquola, C., Shishegar, R., & Fornito, A. (2022). Modeling spatial, developmental, physiological, and topological constraints on human brain connectivity. *Science Advances*, *8*(22), eabm6127. https://doi.org/10.1126/sciadv.abm6127, PubMed: 35658036

Oz, G., Seaquist, E. R., Kumar, A., Criego, A. B., Benedict, L. E., Rao, J. P., … Gruetter, R. (2007). Human brain glycogen content and metabolism: Implications on its role in brain energy metabolism. *American Journal of Physiology-Endocrinology and Metabolism*, *292*(3), E946–E951. https://doi.org/10.1152/ajpendo.00424.2006, PubMed: 17132822

Palmer, T. D., Willhoite, A. R., & Gage, F. H. (2000). Vascular niche for adult hippocampal neurogenesis. *Journal of Comparative Neurology*, *425*(4), 479–494. https://doi.org/10.1002/1096-9861(20001002)425:4<479::aid-cne2>3.0.co;2-3, PubMed: 10975875

Palombo, M., Shemesh, N., Ronen, I., & Valette, J. (2018). Insights into brain microstructure from in vivo DW-MRS. *NeuroImage*, *182*, 97–116. https://doi.org/10.1016/j.neuroimage.2017.11.028, PubMed: 29155183

Pandit, A. S., Expert, P., Lambiotte, R., Bonnelle, V., Leech, R., Turkheimer, F. E., & Sharp, D. J. (2013). Traumatic brain injury impairs small-world topology. *Neurology*, *80*(20), 1826–1833. https://doi.org/10.1212/WNL.0b013e3182929f38, PubMed: 23596068

Papo, D., & Buldú, J. M. (2024). Does the brain behave like a (complex) network? I. Dynamics. *Physics of Life Reviews*, *48*, 47–98. https://doi.org/10.1016/j.plrev.2023.12.006, PubMed: 38145591

Páscoa dos Santos, F., & Verschure, P. F. (2024). Metastable dynamics emerge from local excitatory-inhibitory homeostasis in the cortex at rest. *bioRxiv*. https://doi.org/10.1101/2024.08.24.609501

Peyrounette, M., Davit, Y., Quintard, M., & Lorthois, S. (2018). Multiscale modelling of blood flow in cerebral microcirculation: Details at capillary scale control accuracy at the level of the cortex. *PLoS One*, *13*(1), e0189474. https://doi.org/10.1371/journal.pone.0189474, PubMed: 29324784

Pla, R., Stanco, A., Howard, M. A., Rubin, A. N., Vogt, D., Mortimer, N., … Rubenstein, J. L. R. (2018). Dlx1 and Dlx2 promote interneuron GABA synthesis, synaptogenesis, and dendritogenesis. *Cerebral Cortex*, *28*(11), 3797–3815. https://doi.org/10.1093/cercor/bhx241, PubMed: 29028947

Podvalny, E., Noy, N., Harel, M., Bickel, S., Chechik, G., Schroeder, C. E., … Malach, R. (2015). A unifying principle underlying the extracellular field potential spectral responses in the human cortex. *Journal of Neurophysiology*, *114*(1), 505–519. https://doi.org/10.1152/jn.00943.2014, PubMed: 25855698

Poil, S. S., Hardstone, R., Mansvelder, H. D., & Linkenkaer-Hansen, K. (2012). Critical-state dynamics of avalanches and oscillations jointly emerge from balanced excitation/inhibition in neuronal networks. *Journal of Neuroscience*, *32*(29), 9817–9823. https://doi.org/10.1523/JNEUROSCI.5990-11.2012, PubMed: 22815496

Pope, M., Fukushima, M., Betzel, R. F., & Sporns, O. (2021). Modular origins of high-amplitude cofluctuations in fine-scale functional connectivity dynamics. *Proceedings of the National Academy of Sciences of the United States of America*, *118*(46), e2109380118. https://doi.org/10.1073/pnas.2109380118, PubMed: 34750261

Postnov, D. E., Ryazanova, L. S., & Sosnovtseva, O. V. (2007). Functional modeling of neural-glial interaction. *Biosystems*, *89*(1–3), 84–91. https://doi.org/10.1016/j.biosystems.2006.04.012, PubMed: 17320272

Pulvermuller, F., Tomasello, R., Henningsen-Schomers, M. R., & Wennekers, T. (2021). Biological constraints on neural network models of cognitive function. *Nature Reviews Neuroscience*, *22*(8), 488–502. https://doi.org/10.1038/s41583-021-00473-5, PubMed: 34183826

Rae, C. D. (2014). A guide to the metabolic pathways and function of metabolites observed in human brain 1H magnetic resonance spectra. *Neurochemical Research*, *39*(1), 1–36. https://doi.org/10.1007/s11064-013-1199-5, PubMed: 24258018

Ramezanian-Panahi, M., Abrevaya, G., Gagnon-Audet, J. C., Voleti, V., Rish, I., & Dumas, G. (2022). Generative models of brain dynamics. *Frontiers in Artificial Intelligence*, *5*, 807406. https://doi.org/10.3389/frai.2022.807406, PubMed: 35910192

Ramón y Cajal, S. (1899). *Textura del sistema nervioso del hombre y de los vertebrados : estudios sobre el plan estructural y composición histológica de los centros nerviosos adicionados de consideraciones fisiológicas fundadas en los nuevos descubrimientos* (pp. 1899–1904). Madrid: N. Moya. https://search.library.wisc.edu/catalog/999548808902121

Roca, C., & Adams, R. H. (2007). Regulation of vascular morphogenesis by Notch signaling. *Genes & Development*, *21*(20), 2511–2524. https://doi.org/10.1101/gad.1589207, PubMed: 17938237

Rothman, D. L., Behar, K. L., Hyder, F., & Shulman, R. G. (2003). In vivo NMR studies of the glutamate neurotransmitter flux and neuroenergetics: Implications for brain function. *Annual Review of Physiology*, *65*, 401–427. https://doi.org/10.1146/annurev.physiol.65.092101.142131, PubMed: 12524459

Rowitch, D. H., & Kriegstein, A. R. (2010). Developmental genetics of vertebrate glial–cell specification. *Nature*, *468*(7321), 214–222. https://doi.org/10.1038/nature09611, PubMed: 21068830

Rubinov, M. (2024). Distinctions between brain structure, complexity, and function. Comment on "Does the brain behave like a (complex) network? I. Dynamics" by Papo and Buldú. *Physics of Life Reviews*, *50*, 30–31. https://doi.org/10.1016/j.plrev.2024.05.006, PubMed: 38823364

Sahara, S., Yanagawa, Y., O'Leary, D. D., & Stevens, C. F. (2012). The fraction of cortical GABAergic neurons is constant from near the start of cortical neurogenesis to adulthood. *Journal of Neuroscience*, *32*(14), 4755–4761. https://doi.org/10.1523/JNEUROSCI.6412-11.2012, PubMed: 22492031

Santello, M., Bezzi, P., & Volterra, A. (2011). TNFalpha controls glutamatergic gliotransmission in the hippocampal dentate gyrus. *Neuron*, *69*(5), 988–1001. https://doi.org/10.1016/j.neuron.2011.02.003, PubMed: 21382557

Santello, M., Cali, C., & Bezzi, P. (2012). Gliotransmission and the tripartite synapse. *Advances in Experimental Medicine and Biology*, *970*, 307–331. https://doi.org/10.1007/978-3-7091-0932-8_14, PubMed: 22351062

Sarabi, M. S., Ma, S. J., Jann, K., Ringman, J. M., Wang, D. J. J., & Shi, Y. (2024). Vessel density mapping of small cerebral vessels on 3D high resolution black blood MRI. *NeuroImage*, *286*, 120504. https://doi.org/10.1016/j.neuroimage.2023.120504, PubMed: 38216104

Sertbas, M., & Ulgen, K. O. (2018). Unlocking human brain metabolism by genome-scale and multiomics metabolic models: Relevance for neurology research, health, and disease. *OMICS*, *22*(7), 455–467. https://doi.org/10.1089/omi.2018.0088, PubMed: 30004841

Shanahan, M. (2010). Metastable chimera states in community-structured oscillator networks. *Chaos*, *20*(1), 013108. https://doi.org/10.1063/1.3305451, PubMed: 20370263

Shipley, R. J., Smith, A. F., Sweeney, P. W., Pries, A. R., & Secomb, T. W. (2020). A hybrid discrete-continuum approach for modelling microcirculatory blood flow. *Mathematical Medicine and Biology*, *37*(1), 40–57. https://doi.org/10.1093/imammb/dqz006, PubMed: 30892609

Shonkoff, J. P., Phillips, D. A., & National Research Council (U.S.). Committee on Integrating the Science of Early Childhood Development. (2000). *From neurons to neighborhoods: The science of early child development*. National Academy Press.

Shu, Y., Hasenstaub, A., & McCormick, D. A. (2003). Turning on and off recurrent balanced cortical activity. *Nature*, *423*(6937), 288–293. https://doi.org/10.1038/nature01616, PubMed: 12748642

Smith, T. G., Jr., Lange, G. D., & Marks, W. B. (1996). Fractal methods and results in cellular morphology—Dimensions, lacunarity and multifractals. *Journal of Neuroscience Methods*, *69*(2), 123–136. https://doi.org/10.1016/S0165-0270(96)00080-5, PubMed: 17293862

Smith, S. L., & Trachtenberg, J. T. (2007). Experience-dependent binocular competition in the visual cortex begins at eye opening. *Nature Neuroscience*, *10*(3), 370–375. https://doi.org/10.1038/nn1844, PubMed: 8946315

Southwell, D. G., Paredes, M. F., Galvao, R. P., Jones, D. L., Froemke, R. C., Sebe, J. Y., ... Alvarez-Buylla, A. (2012). Intrinsically determined cell death of developing cortical interneurons. *Nature*, *491*(7422), 109–113. https://doi.org/10.1038/nature11523, PubMed: 23041929

Sporns, O. (2022). The complex brain: Connectivity, dynamics, information. *Trends in Cognitive Sciences*, *26*(12), 1066–1067. https://doi.org/10.1016/j.tics.2022.08.002, PubMed: 36207260

Sporns, O., & Kötter, R. (2004). Motifs in brain networks. *PLoS Biology*, *2*(11), e369. https://doi.org/10.1371/journal.pbio.0020369, PubMed: 15510229

Stam, C. J., Jones, B. F., Nolte, G., Breakspear, M., & Scheltens, P. (2007). Small-world networks and functional connectivity in Alzheimer's disease. *Cerebral Cortex*, *17*(1), 92–99. https://doi.org/10.1093/cercor/bhj127, PubMed: 16452642

Strogatz, S. H. (2001). Exploring complex networks. *Nature*, *410*(6825), 268–276. https://doi.org/10.1038/35065725, PubMed: 11258382

Takeda, T., Ishikawa, A., Ohtomo, K., Kobayashi, Y., & Matsuoka, T. (1992). Fractal dimension of dendritic tree of cerebellar Purkinje cell during onto- and phylogenetic development. *Neuroscience Research*, *13*(1), 19–31. https://doi.org/10.1016/0168-0102(92)90031-7, PubMed: 1314350

Tononi, G., Edelman, G. M., & Sporns, O. (1998). Complexity and coherency: Integrating information in the brain. *Trends in Cognitive Sciences*, *2*(12), 474–484. https://doi.org/10.1016/S1364-6613(98)01259-5, PubMed: 21227298

Touboul, J., Wendling, F., Chauvel, P., & Faugeras, O. (2011). Neural mass activity, bifurcations, and epilepsy. *Neural Computation*, *23*(12), 3232–3286. https://doi.org/10.1162/NECO_a_00206, PubMed: 21919787

Turkheimer, F. E., Leech, R., Expert, P., Lord, L.-D., & Vernon, A. C. (2015). The brain's code and its canonical computational motifs. From sensory cortex to the default mode network: A multi-scale model of brain function in health and disease. *Neuroscience and Biobehavioral Reviews*, *55*, 211–222. https://doi.org/10.1016/j.neubiorev.2015.04.014, PubMed: 25956253

Turkheimer, F. E., Rosas, F. E., Dipasquale, O., Martins, D., Fagerholm, E. D., Expert, P., ... Leech, R. (2022). A complex systems perspective on neuroimaging studies of behavior and its disorders.

Neuroscientist, *28*(4), 382–399. https://doi.org/10.1177/1073858421994784, PubMed: 33593120

Van Ooyen, A., Van Pelt, J., & Corner, M. A. (1995). Implications of activity dependent neurite outgrowth for neuronal morphology and network development. *Journal of Theoretical Biology*, *172*(1), 63–82. https://doi.org/10.1006/jtbi.1995.0005, PubMed: 7891450

van Vreeswijk, C., & Sompolinsky, H. (1996). Chaos in neuronal networks with balanced excitatory and inhibitory activity. *Science*, *274*(5293), 1724–1726. https://doi.org/10.1126/science.274.5293.1724, PubMed: 8939866

van Vreeswijk, C., & Sompolinsky, H. (1998). Chaotic balanced state in a model of cortical circuits. *Neural Computation*, *10*(6), 1321–1371. https://doi.org/10.1162/089976698300017214, PubMed: 9698348

Váša, F., Shanahan, M., Hellyer, P. J., Scott, G., Cabral, J., & Leech, R. (2015). Effects of lesions on synchrony and metastability in cortical networks. *NeuroImage*, *118*, 456–467. https://doi.org/10.1016/j.neuroimage.2015.05.042, PubMed: 26049146

Verkhratsky, A., & Nedergaard, M. (2014). Astroglial cradle in the life of the synapse. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *369*(1654), 20130595. https://doi.org/10.1098/rstb.2013.0595, PubMed: 25225089

Verkhratsky, A., & Nedergaard, M. (2018). Physiology of astroglia. *Physiological Reviews*, *98*(1), 239–389. https://doi.org/10.1152/physrev.00042.2016, PubMed: 29351512

Vértes, P. E. (2023). Computational models of typical and atypical brain network development. *Biological Psychiatry*, *93*(5), 464–470. https://doi.org/10.1016/j.biopsych.2022.11.012, PubMed: 36593135

Vértes, P. E., Alexander-Bloch, A. F., Gogtay, N., Giedd, J. N., Rapoport, J. L., & Bullmore, E. T. (2012). Simple models of human brain functional networks. *Proceedings of the National Academy of Sciences of the United States of America*, *109*(15), 5868–5873. https://doi.org/10.1073/pnas.1111738109, PubMed: 22467830

Vogels, T. P., Sprekeler, H., Zenke, F., Clopath, C., & Gerstner, W. (2011). Inhibitory plasticity balances excitation and inhibition in sensory pathways and memory networks. *Science*, *334*(6062), 1569–1573. https://doi.org/10.1126/science.1211095, PubMed: 22075724

Wang, X. J., & Buzsáki, G. (1996). Gamma oscillation by synaptic inhibition in a hippocampal interneuronal network model. *Journal of Neuroscience*, *16*(20), 6402–6413. https://doi.org/10.1523/JNEUROSCI.16-20-06402.1996, PubMed: 8815919

Wang, Y., Yu, S., & Li, M. (2023). Neurovascular crosstalk and cerebrovascular alterations: An underestimated therapeutic target in autism spectrum disorders. *Frontiers in Cellular Neuroscience*, *17*, 1226580. https://doi.org/10.3389/fncel.2023.1226580, PubMed: 37692552

Wang, Z., Ying, Z., Bosy-Westphal, A., Zhang, J., Schautz, B., Later, W., ... Müller, M. J. (2010). Specific metabolic rates of major organs and tissues across adulthood: Evaluation by mechanistic model of resting energy expenditure. *American Journal of Clinical Nutrition*, *92*(6), 1369–1377. https://doi.org/10.3945/ajcn.2010.29885, PubMed: 20962155

Waschke, L., Donoghue, T., Fiedler, L., Smith, S., Garrett, D. D., Voytek, B., & Obleser, J. (2021). Modality-specific tracking of attention and sensory statistics in the human electrophysiological spectral exponent. *eLife*, *10*, e70068. https://doi.org/10.7554/eLife.70068, PubMed: 34672259

Watts, D. J., & Strogatz, S. H. (1998). Collective dynamics of 'small-world' networks. *Nature*, *393*(6684), 440–442. https://doi.org/10.1038/30918, PubMed: 9623998

Wearne, S. L., Rodriguez, A., Ehlenberger, D. B., Rocher, A. B., Henderson, S. C., & Hof, P. R. (2005). New techniques for imaging, digitization and analysis of three-dimensional neural morphology on multiple scales. *Neuroscience*, *136*(3), 661–680. https://doi.org/10.1016/j.neuroscience.2005.05.053, PubMed: 16344143

Wigström, H., & Gustafsson, B. (1985). Facilitation of hippocampal long-lasting potentiation by GABA antagonists. *Acta Physiologica Scandinavica*, *125*(1), 159–172. https://doi.org/10.1111/j.1748-1716.1985.tb07703.x, PubMed: 2996303

Wu, G. K., Tao, H. W., & Zhang, L. I. (2011). From elementary synaptic circuits to information processing in primary auditory cortex. *Neuroscience and Biobehavioral Reviews*, *35*(10), 2094–2104. https://doi.org/10.1016/j.neubiorev.2011.05.004, PubMed: 21609731

Wu, M. W., Kourdougli, N., & Portera-Cailliau, C. (2024). Network state transitions during cortical development. *Nature Reviews Neuroscience*, *25*(8), 535–552. https://doi.org/10.1038/s41583-024-00824-y, PubMed: 38783147

Xie, T., Stathopoulou, M. G., de Andrés, F., Siest, G., Murray, H., Martin, M., ... Visvikis-Siest, S. (2017). VEGF-related polymorphisms identified by GWAS and risk for major depression. *Translational Psychiatry*, *7*(3), e1055. https://doi.org/10.1038/tp.2017.36, PubMed: 28267147

Yang, A. C., Vest, R. T., Kern, F., Lee, D. P., Agam, M., Maat, C. A., ... Wyss-Coray, T. (2022). A human brain vascular atlas reveals diverse mediators of Alzheimer's risk. *Nature*, *603*(7903), 885–892. https://doi.org/10.1038/s41586-021-04369-3, PubMed: 35165441

Yang, X., & La Camera, G. (2024). Co-existence of synaptic plasticity and metastable dynamics in a spiking model of cortical circuits. *PLoS Computational Biology*, *20*(7), e1012220. https://doi.org/10.1371/journal.pcbi.1012220, PubMed: 38950068

Yang, Y., Guan, W., Sheng, X.-M., & Gu, H.-J. (2024). Role of Semaphorin 3A in common psychiatric illnesses such as schizophrenia, depression, and anxiety. *Biochemical Pharmacology*, *226*, 116358. https://doi.org/10.1016/j.bcp.2024.116358, PubMed: 38857830

Yarkoni, T., Poldrack, R. A., Nichols, T. E., Van Essen, D. C., & Wager, T. D. (2011). Large-scale automated synthesis of human functional neuroimaging data. *Nature Methods*, *8*(8), 665–670. https://doi.org/10.1038/nmeth.1635, PubMed: 21706013

Ye, Z., Mostajo-Radji, M. A., Brown, J. R., Rouaux, C., Tomassy, G. S., Hensch, T. K., & Arlotta, P. (2015). Instructing perisomatic inhibition by direct lineage reprogramming of neocortical projection neurons. *Neuron*, *88*(3), 475–483. https://doi.org/10.1016/j.neuron.2015.10.006, PubMed: 26539889

Zang, Y., Chaudhari, K., & Bashaw, G. J. (2021). New insights into the molecular mechanisms of axon guidance receptor regulation and signaling. *Current Topics in Developmental Biology*, *142*, 147–196. https://doi.org/10.1016/bs.ctdb.2020.11.008, PubMed: 33706917

Zhang, S., Larsen, B., Sydnor, V. J., Zeng, T., An, L., Yan, X., ... Thomas Yeo, B. T. (2024). In-vivo whole-cortex marker of excitation-inhibition ratio indexes cortical maturation and cognitive ability in youth. *bioRxiv*. https://doi.org/10.1101/2023.06.22.546023, PubMed: 38586012

Zhang, X., Braun, U., Harneit, A., Zang, Z., Geiger, L. S., Betzel, R. F., ... Tost, H. (2021). Generative network models of altered structural brain connectivity in schizophrenia. *NeuroImage*, *225*, 117510. https://doi.org/10.1016/j.neuroimage.2020.117510, PubMed: 33160087